



Data Exfiltration Investigation Report

Investigation profile: Data Exfiltration (DLP alert)

Data leakage - INFORMANT-PC

Case number	CFREDS-2015-DL
Case name	Data leakage - INFORMANT-PC
Client / requester	-
Examiner	Examiner
Organization	Forensic Lab
Report date	07 October 2026
Evidence items	Suspect workstation (INFORMANT-PC)
Time zone	All times UTC; local times shown in America/New_York

This report was produced with the assistance of automated analysis. All findings were generated from the examined evidence and must be reviewed and validated by the examiner before the report is relied upon.

Document control

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1. Executive summary

Forensic Lab examined 1 evidence item(s) - Suspect workstation (INFORMANT-PC) - using the 'Data Exfiltration (DLP alert)' investigation profile of Windows Forensic Automation. Background: Employee Iaman Informant is suspected of leaking confidential project documents to an outside party (spy.conspirator) via e-mail, cloud storage and removable media, and of attempting to destroy evidence.

Evidence examined

Evidence	Role	Host	Operating system	Size	Integrity
Suspect workstation	Corporate laptop / workstation (data source)	INFORMANT-PC	Windows 7 Ultimate (7601)	21.5 GB	container checks only

Key conclusions

- **Q1 - Yes:** Was removable storage connected to the examined system(s)? When, and by which user?
 - Suspect workstation (INFORMANT-PC): SanDisk Cruzer Fit S/N 4C530012450531101593 (first connected 2015-03-23 18:31:09 UTC, connected after the last reboot 2015-03-24 13:38:00 UTC, user informant).
 - Suspect workstation (INFORMANT-PC): SanDisk Cruzer Fit S/N 4C530012550531106501 (first connected 2015-03-24 13:58:31 UTC, connected after the last reboot 2015-03-24 13:58:33 UTC).
- **Q4 - Indicated:** Is there evidence that the files were copied to, or opened from, removable media?
 - Suspect workstation (INFORMANT-PC): 11 shortcut/jump-list records of files on removable media.
 - Suspect workstation (INFORMANT-PC): 19 shellbag/MRU/Timeline records reference removable drives.
- **Q5 - Indicated:** Is there evidence that the files were uploaded via webmail, cloud storage or file-sharing websites?
 - Suspect workstation (INFORMANT-PC): browser file dialogs were used.
- **Q6 - Indicated:** Were other channels used (e-mail, cloud sync client, CD/DVD burning, printing, archiving / transfer tools)?
 - Suspect workstation (INFORMANT-PC): e-mail correspondence with spy.conspirator@nist.gov (15), iaman.informant@nist.gov (14).
 - Suspect workstation (INFORMANT-PC): 2 file upload(s) by the Google Drive (legacy) client (do_u_wanna_build_a_snow_man.mp3, happy_holiday.jpg).
 - Suspect workstation (INFORMANT-PC): The cdrom driver logged 4 completed disc write(s) (System event 133) between 2015-03-24 19:47:47 and 2015-03-24 20:41:21 UTC. \$UsnJrnl records 8 disc-mastering session(s) (IMAPI DAT/FIL/POST temporary files). 62 file(s) were placed in the Explorer burn staging folder (AppData\Local\Microsoft\Windows\Burn\Burn).
 - Suspect workstation (INFORMANT-PC): Google Drive / Backup and Sync (legacy client) executed.
- **Q9 - Yes:** Is there evidence of attempts to conceal the activity or remove traces?
 - Suspect workstation (INFORMANT-PC): CCleaner; Eraser; Search / page about hiding traces; Secure deletion pattern.
- **Q10 - Yes:** Do residual data areas (unallocated space, slack, pagefile, hibernation file) contain traces of the files or activity?
 - Suspect workstation (INFORMANT-PC): 13 relevant strings in memory files.
 - Suspect workstation (INFORMANT-PC): hits in \$I30 directory index (incl. index slack), \$LogFile (NTFS transaction log), \$UsnJrnl (change journal), File slack, MFT record (resident data / metadata), Unallocated (deleted file remnant), Unallocated space, Volume Shadow Copy store, pagefile.sys.

Investigative questions and conclusions

#	Question	Conclusion	Basis
Q1	Was removable storage connected to the examined system(s)? When, and by which user?	Yes	Suspect workstation (INFORMANT-PC): SanDisk Cruzer Fit S/N 4C530012450531101593 (first connected 2015-03-23 18:31:09 UTC, connected after the last reboot 2015-03-24 13:38:00 UTC, user informant). Suspect workstation (INFORMANT-PC): SanDisk Cruzer Fit S/N 4C530012550531106501 (first connected 2015-03-24 13:58:31 UTC, connected after the last reboot 2015-03-24 13:58:33 UTC). (See F-04, F-05)
Q2	Was the removable device reported in the DLP alert connected to the examined system(s)?	Not applicable	No removable media serial was supplied.
Q3	Were the files of interest present on the examined system(s)? Were they deleted, renamed or archived?	Not applicable	No files of interest were supplied; see the document access findings.
Q4	Is there evidence that the files were copied to, or opened from, removable media?	Indicated	Suspect workstation (INFORMANT-PC): 11 shortcut/jump-list records of files on removable media. Suspect workstation (INFORMANT-PC): 19 shellbag/MRU/Timeline records reference removable drives. (See F-06)
Q5	Is there evidence that the files were uploaded via webmail, cloud storage or file-sharing websites?	Indicated	Suspect workstation (INFORMANT-PC): browser file dialogs were used. (See F-08)
Q6	Were other channels used (e-mail, cloud sync client, CD/DVD burning, printing, archiving / transfer tools)?	Indicated	Suspect workstation (INFORMANT-PC): e-mail correspondence with spy.conspirator@nist.gov (15), iaman.informant@nist.gov (14). Suspect workstation (INFORMANT-PC): 2 file upload(s) by the Google Drive (legacy) client (do_u_wanna_build_a_snow_man.mp3, happy_holiday.jpg). Suspect workstation (INFORMANT-PC): The cdrom driver logged 4 completed disc write(s) (System event 133) between 2015-03-24 19:47:47 and 2015-03-24 20:41:21 UTC. \$UsnJrnl records 8 disc-mastering session(s) (IMAPI DAT/FIL/POST temporary files). 62 file(s) were placed in the Explorer burn staging folder (AppData\Local\Microsoft\Windows\Burn\Burn). Suspect workstation (INFORMANT-PC): Google Drive / Backup and Sync (legacy client) (See F-09, F-10, F-11, F-12)
Q7	Did the files reach the personal / secondary device, and how?	Not applicable	No personal / secondary device was examined.
Q8	Do the system artifacts corroborate each DLP alert?	Not applicable	No DLP alert export was supplied.
Q9	Is there evidence of attempts to conceal the activity or remove traces?	Yes	Suspect workstation (INFORMANT-PC): CCleaner; Eraser; Search / page about hiding traces; Secure deletion pattern. (See F-03)
Q10	Do residual data areas (unallocated space, slack, pagefile, hibernation file) contain traces of the files or activity?	Yes	Suspect workstation (INFORMANT-PC): 13 relevant strings in memory files. Suspect workstation (INFORMANT-PC): hits in \$!30 directory index (incl. index slack), \$LogFile (NTFS transaction log), \$UsnJrnl (change journal), File slack, MFT record (resident data / metadata), Unallocated (deleted file remnant), Unallocated space, Volume Shadow Copy store, pagefile.sys. (See F-13, F-15)

Conclusion key: 'Yes' - established by the artifacts; 'Indicated' - supported by artifacts but not conclusive on its own; 'No evidence found' - the relevant artifacts were examined and nothing was found (absence of evidence is not evidence of absence - see the coverage matrix in section 8); 'Not applicable' - the question could not apply to the evidence or inputs supplied.

2. Case background and examiner inputs

Employee Iaman Informant is suspected of leaking confidential project documents to an outside party (spy.conspirator) via e-mail, cloud storage and removable media, and of attempting to destroy evidence.

2.1 Scope

Profile: Data Exfiltration (DLP alert). Use when a DLP solution (Purview, Symantec, Forcepoint, Netskope, ...) or a manager reports that an employee copied or uploaded files. Load the corporate laptop and, if available, the personal laptop and the USB device. Every input is optional - the investigation adapts to whatever you know: with a DLP export the alerts are corroborated one by one; with only file names / hashes or reference copies the files are traced everywhere; with nothing but a time window the removable media and web activity in that window is reconstructed.

2.2 Other inputs

Search terms	3 terms: secret_project, spy.conspirator, confidential
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3. Evidence examined

Each evidence item was opened read-only. Container metadata and, where requested, a full verification hash were recorded. The operating system details below were read from the image itself.

3.1 Suspect workstation (INFORMANT-PC)

Label / role	Suspect workstation - Corporate laptop / workstation (data source)
Image file(s)	C:\Users\sujay\Desktop\data-leackagecase\cfreds_2015_data_leakage_pc.E01 (4 segments)
Format	ewf
Media size	21.47 GB
Acquisition metadata	case_number: 0x11; description: cfreds_2015_data_leakage_pc; examiner_name: dForensics_Team; evidence_number: 0x01; acquiry_date: Thu Apr 23 20:28:22 2015; acquiry_software_version: 7.10
Acquisition hash	MD5 a49d1254c873808c58e6f1bcd60b5bde, SHA1 afe5c9ab487bd47a8a9856b1371c2384d44fd785
Verification	Not re-verified during this examination
Host name	INFORMANT-PC
Operating system	Windows 7 Ultimate (build 7601)
Installed	2015-03-22 14:34:26 UTC
Time zone	Eastern Standard Time (America/New_York)
Last shutdown	2015-03-25 15:31:05 UTC
User profiles	systemprofile, LocalService, NetworkService, informant, admin11, temporary
Volumes	part_00100000: NTFS 'System Reserved' serial 4A180A15180A0125 (0.10 GB) part_06500000: NTFS " serial C8CA0C8DCA0C7A48 (21.37 GB)

4. Methodology

The evidence was processed with Windows Forensic Automation 1.2.0, an automated, profile-driven Windows forensic platform. Each image was opened read-only; no data was written to the evidence. The following steps were performed for every Windows system image:

- **Web browsers:** History, downloads, search terms, saved-login user names, autofill entries, cookie hosts and extensions for Chrome, Edge, Brave, Opera, Vivaldi, Firefox and other Chromium/Gecko browsers.
- **Cloud storage clients:** OneDrive, Google Drive for desktop, Dropbox, Box, MEGA, iCloud, pCloud, Nextcloud: signed-in accounts, local sync folders and synced item databases.

- **Raw artifact collection (for manual analysis):** Copies registry hives (+ transaction logs), \$MFT, \$LogFile, \$UsnJrnl:\$J, event logs, Prefetch, SRUM, Amcache, LNK / jump lists, browser databases, e-mail stores, scheduled tasks, setupapi logs, WMI repository, PowerShell history and more into the case folder, hashing every file into a manifest.
- **Documents (metadata + full text):** Office/ODF/PDF/RTF/text documents in user profiles and on non-system volumes: author / last saved by / internal timestamps / macros, SHA-256, and full text into the case search index.
- **E-mail stores:** Outlook PST/OST mailboxes, .eml and .msg files: message metadata and every attachment (name, size, MD5/SHA-256).
- **Windows event logs:** Parses every EVTX log (Security, System, Application, PowerShell, RDP, Task Scheduler, Defender, Sysmon, WLAN, Print, Partition, Kernel-PnP, ...) and keeps forensically relevant events.
- **Program execution:** Prefetch (run times, files and volumes referenced), Amcache, PCA, Shimcache, BAM/DAM, SRUM, MUICache.
- **Shortcuts, jump lists, Recycle Bin, Timeline:** LNK shortcut files (incl. deleted / resident ones), jump lists (automatic + custom destinations), Recycle Bin \$I/\$R pairs and the Windows Timeline (ActivitiesCache.db).
- **File system index (MFT / FAT / USN):** Indexes every file record including deleted ones, alternate data streams (download origin), timestamp anomalies, the USN change journal and a cluster map for slack/unallocated attribution.
- **Internet Explorer / Edge (legacy) history:** WebCacheV01.dat (IE 10/11, Edge legacy) and index.dat (IE 5-9) history and downloads, including file:/// records of documents opened from local, removable and network drives.
- **Pagefile / hiberfil / swapfile strings:** Streams pagefile.sys, swapfile.sys, hiberfil.sys and crash dumps and extracts URLs, e-mail addresses and document paths, keeping webmail / cloud / file-transfer / remote-access destinations, case keywords and target file names.
- **Network configuration:** Wi-Fi profiles (SSID, authentication), hosts file entries, proxy / PAC settings, VPN phonebooks.
- **CD / DVD burning:** Files staged for disc burning (allocated and deleted), CD Burning registry state per user, optical volumes.
- **Persistence mechanisms:** Run/RunOnce keys, services, scheduled tasks (XML), startup folders, Winlogon shell/userinit, IFEO debuggers, Applnit DLLs and WMI event subscriptions - each command scored for suspicious traits.
- **Physical keyword search & carving:** Every keyword is searched across every byte of every volume (allocated files, file slack, unallocated clusters, MFT, \$LogFile, USN journal, directory indexes, pagefile/hiberfil/swapfile, shadow copy stores) and unpartitioned space, in ASCII/UTF-8 and UTF-16LE. Deleted documents, shortcuts, databases and executables are carved from unallocated space. Keywords are also matched against document text.
- **Remote access & RMM tools:** Detects AnyDesk, TeamViewer, ScreenConnect, Splashtop, Atera, RustDesk, NetSupport, VNC and 20+ other remote tools from files, services, installs and execution traces, and parses their connection logs.
- **PowerShell history & scripts:** PSReadLine console history per user, PowerShell transcripts, and script / container files (ps1, vbs, js, hta, bat, iso, msi, Ink...) in user-writable folders, each scored for malicious traits.
- **Sticky Notes:** Text of Windows Sticky Notes with creation / modification times (StickyNotes.snt, plum.sqlite).
- **System information & accounts:** OS build, time zone, shutdown time, local accounts (SAM), network profiles, interfaces, installed software.
- **Target file search (hash / name / content):** Looks for every file of interest supplied by the examiner (DLP export, hash list, reference copies) in allocated and deleted files, the Recycle Bin, carved files, e-mail attachments, ZIP archives and Volume Shadow Copies, by hash, name, size, text similarity and document metadata.
- **USB & removable media:** Builds the removable device history (USB mass storage, UAS/SCSI, MTP phones, SD) with first/last connection times, drive letters, volume serials/labels, users and connection sessions.
- **User activity (registry):** RecentDocs, Open/Save and LastVisited MRU, Run box (Win+R) history, typed paths, Explorer searches, UserAssist, Office file MRU, trusted documents, typed URLs, RDP client MRU and shellbags.
- **Windows Search index:** Windows.edb (Windows Search): indexed files, folders, Outlook e-mails and web shortcuts, including items that were later deleted.

4.1 Search methodology

Physical keyword search reads every byte of every volume and of unpartitioned disk space and matches every term in ASCII/UTF-8 and UTF-16LE, case-insensitively. Each hit is attributed using the file system's own allocation structures to an allocated file, file slack, a deleted file's former clusters, unallocated space, an MFT record, the \$LogFile, the USN journal, a directory index, pagefile.sys / hiberfil.sys / swapfile.sys or a shadow copy store. Because the content of compressed documents (DOCX, XLSX, PDF) is not visible to a physical search, the text of every document - including documents recovered from deleted entries and carved from unallocated space - was also indexed and searched. Files of interest were located by MD5/SHA-1/SHA-256, by name and size, by text similarity and by document metadata, in allocated and deleted files, the Recycle Bin, ZIP archives, e-mail attachments, carved files and Volume Shadow Copies.

4.2 Time

All timestamps are stored and reported in UTC; where useful the local time in America/New_York is shown in brackets. Artifacts that record local time (setupapi logs, FAT directory entries, NetworkList dates) were converted using the time zone configured on the source system, or the assumption stated in the limitations.

5. Findings

5.1 Q1: Was removable storage connected to the examined system(s)? When, and by which user?

Conclusion: Yes - Suspect workstation (INFORMANT-PC): SanDisk Cruzer Fit S/N 4C530012450531101593 (first connected 2015-03-23 18:31:09 UTC, connected after the last reboot 2015-03-24 13:38:00 UTC, user informant). Suspect workstation (INFORMANT-PC): SanDisk Cruzer Fit S/N 4C530012550531106501 (first connected 2015-03-24 13:58:31 UTC, connected after the last reboot 2015-03-24 13:58:33 UTC).

F-04 USB device SanDisk Cruzer Fit (S/N 4C530012450531101593) connected to Suspect workstation (INFORMANT-PC)

System: Suspect workstation (INFORMANT-PC)	Time: 2015-03-24 13:38:00 UTC	Supporting artifacts: 2 Removable media
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The device 'SanDisk Cruzer Fit' with serial number 4C530012450531101593 was connected to Suspect workstation (INFORMANT-PC). First connected 2015-03-23 18:31:09 UTC (2015-03-23 14:31:09 UTC-04:00); connected after the last reboot 2015-03-24 13:38:00 UTC (2015-03-24 09:38:00 UTC-04:00) (DeviceClasses key - later reconnections during the same boot session are not recorded there); volume last mounted by a user 2015-03-24 13:38:00 UTC (2015-03-24 09:38:00 UTC-04:00). The volume was mounted by user(s): informant (MountPoints2 volume key linked to this device because it was written within seconds of the device's arrival). Connection sessions: 2015-03-24 13:38:00 (disconnection time not recorded).

Property	Value
Device	SanDisk Cruzer Fit
Device class	USB mass storage
Serial number	4C530012450531101593
VID:PID	0781:5571
Friendly name	SanDisk Cruzer Fit USB Device
First connected (UTC)	2015-03-23 18:31:09
Connected after last reboot (UTC)	2015-03-24 13:38:00
Volume last mounted by user (UTC)	2015-03-24 13:38:00
Connections observed	1
Volume GUID	{662626a6-d181-11e4-9591-000c29ff2429}
Mounted by user(s)	informant
Sources	DeviceClasses, EVTX System 20001, EVTX System 20003, Enum\USB, Enum\USBSTOR, ...

12 rows | 3 highlighted

1 Device serial number

2 User profile that mounted the volume (MountPoints2)

Figure 1: Suspect workstation (INFORMANT-PC) - removable device details. USB device history reconstructed from DeviceClasses, EVTX System 20001, EVTX System 20003, Enum\USB, Enum\USBSTOR, MountPoints2 (time-correlated), setupapi.dev.log

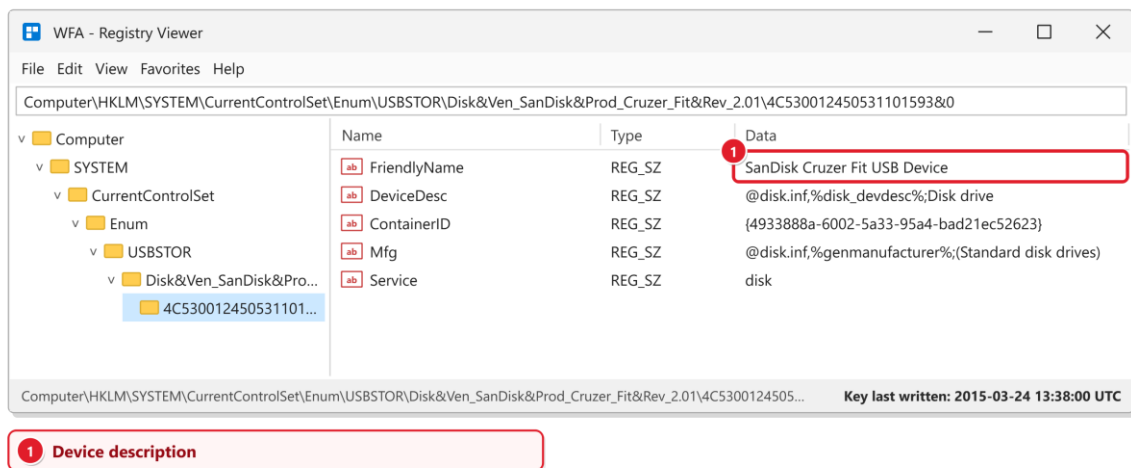


Figure 2: Registry - USBSTOR entry for the device. SYSTEM hive: device enumeration key (no arrival / removal properties are recorded on this Windows version)

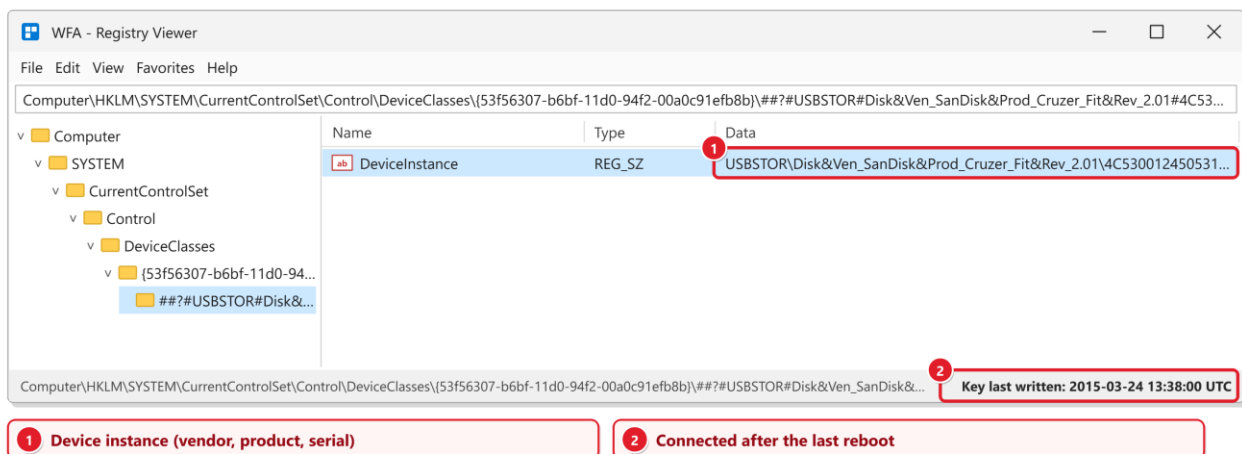


Figure 3: Registry - device interface key (connected after the last reboot). SYSTEM hive: device interface keys are written when the device is first connected after a reboot; later reconnections during the same boot session do not update them

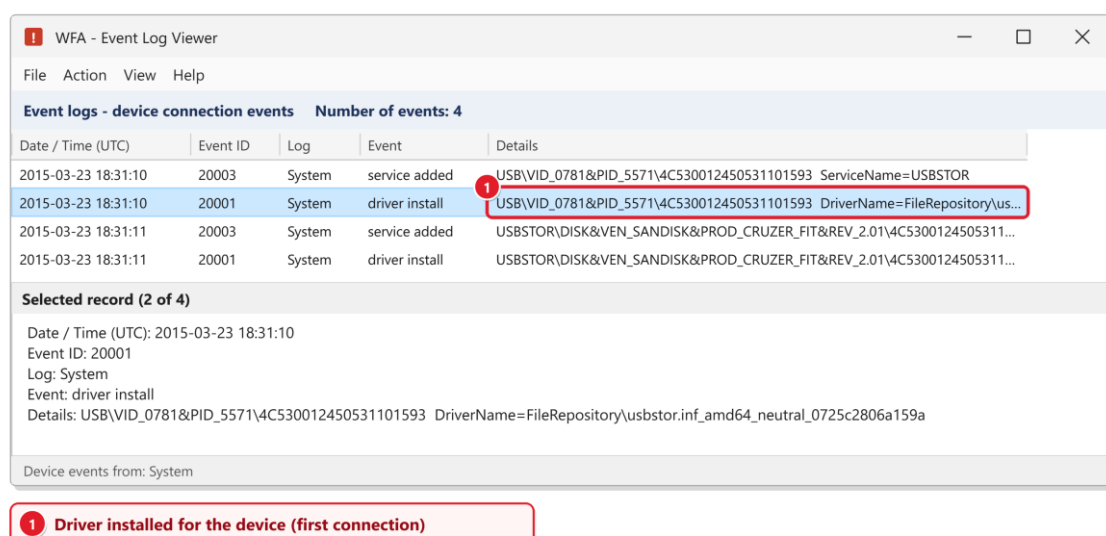


Figure 4: Event logs - device connection events. Device events from: System

Supporting artifacts

Artifact	Time (UTC)	Summary	Source
USB / Removable Devices	2015-03-24 13:38:00	SanDisk Cruzer Fit S/N 4C530012450531101593 first connected 2015-03-23 18:31:09.394000; connected after the last reboot 2015-03-24 13:38:00.034222	DeviceClasses, EVTX System 20001, EVTX System 20003, Enum\USB, Enum\USBSTOR, MountPoints2 (time-correlated), setupapi.de
USB Connection Sessions	2015-03-24 13:38:00	SanDisk Cruzer Fit 4C530012450531101593 connected 2015-03-24 13:38:00.034222 -> unknown	DeviceClasses key last written (first connection after the last reboot)

F-05 USB device SanDisk Cruzer Fit (S/N 4C530012550531106501) connected to Suspect workstation (INFORMANT-PC)

System: Suspect workstation (INFORMANT-PC)	Time: 2015-03-24 13:58:33 UTC	Supporting artifacts: 2	Removable media
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The device 'SanDisk Cruzer Fit' with serial number 4C530012550531106501 was connected to Suspect workstation (INFORMANT-PC). First connected 2015-03-24 13:58:31 UTC (2015-03-24 09:58:31 UTC-04:00); connected after the last reboot 2015-03-24 13:58:33 UTC (2015-03-24 09:58:33 UTC-04:00) (DeviceClasses key - later reconnections during the same boot session are not recorded there). It was assigned drive letter E:. Its volume carried label 'IAMAN \$ _@' and volume serial . Connection sessions: 2015-03-24 13:58:33 (disconnection time not recorded).

WFA - USB & Removable Media

Suspect workstation (INFORMANT-PC) - removable device details

Property	Value
Device	SanDisk Cruzer Fit
Device class	USB mass storage
Serial number	4C530012550531106501
VID:PID	0781:5571
Friendly name	SanDisk Cruzer Fit USB Device
First connected (UTC)	2015-03-24 13:58:31
Connected after last reboot (U...	2015-03-24 13:58:33
Connections observed	1
Drive letter	E:
Volume label	IAMAN \$ _@
Sources	DeviceClasses, EVTX System 20001, EVTX System 20003, Enum\USB, Enum\USBSTOR, V...

11 rows | 2 highlighted

1 Device serial number

Figure 5: Suspect workstation (INFORMANT-PC) - removable device details. USB device history reconstructed from DeviceClasses, EVTX System 20001, EVTX System 20003, Enum\USB, Enum\USBSTOR, VolumeInfoCache (time-correlated), setupapi.dev.log

WFA - Registry Viewer

File Edit View Favorites Help

Computer\HKLMSYSTEM\CurrentControlSet\Enum\USBSTOR\Disk&Ven_SanDisk&Prod_Cruzer_Fit&Rev_2.01\4C530012550531106501&0

Name	Type	Data
FriendlyName	REG_SZ	SanDisk Cruzer Fit USB Device
DeviceDesc	REG_SZ	@disk.inf,%disk_devdesc%;Disk drive
ContainerID	REG_SZ	{9f935160-3dcc-5a21-a9bc-b9311fd83c43}
Mfg	REG_SZ	@disk.inf,%genmanufacturer%;(Standard disk drives)
Service	REG_SZ	disk

Computer\HKLMSYSTEM\CurrentControlSet\Enum\USBSTOR\Disk&Ven_SanDisk&Prod_Cruzer_Fit&Rev_2.01\4C5300125505... Key last written: 2015-03-24 13:58:33 UTC

1 Device description

Figure 6: Registry - USBSTOR entry for the device. SYSTEM hive: device enumeration key (no arrival / removal properties are recorded on this Windows version)

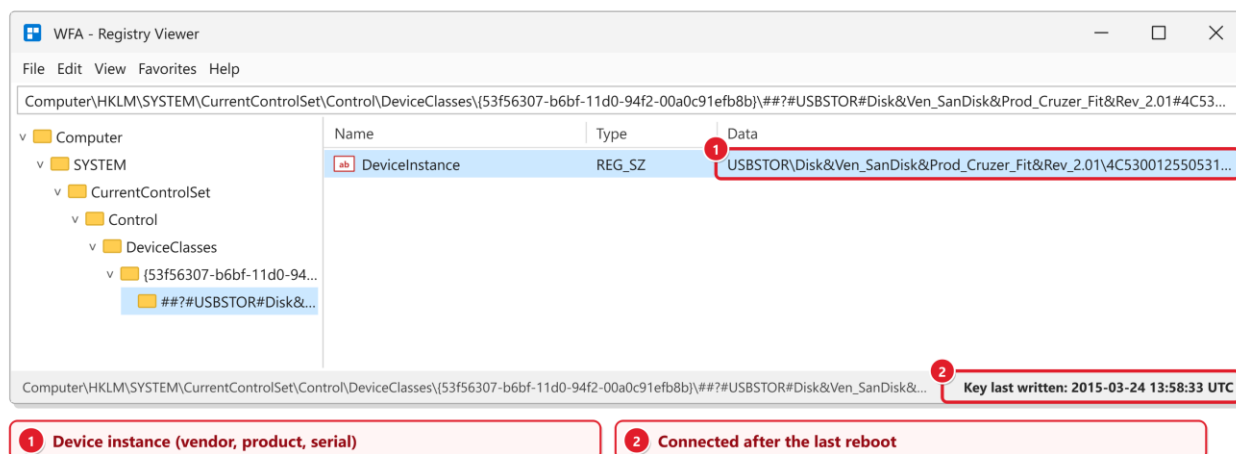


Figure 7: Registry - device interface key (connected after the last reboot). SYSTEM hive: device interface keys are written when the device is first connected after a reboot; later reconnections during the same boot session do not update them

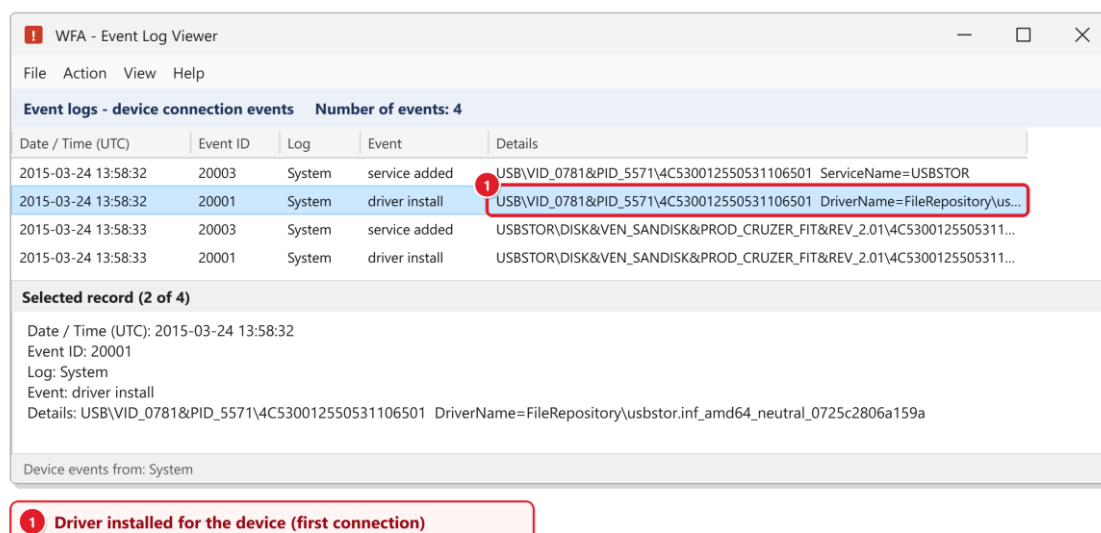


Figure 8: Event logs - device connection events. Device events from: System

Supporting artifacts

Artifact	Time (UTC)	Summary	Source
USB / Removable Devices	2015-03-24 13:58:33	SanDisk Cruzer Fit S/N 4C530012550531106501 first connected 2015-03-24 13:58:31.856000; connected after the last reboot 2015-03-24 13:58:33.518909	DeviceClasses, EVTX System 20001, EVTX System 20003, Enum\USB, Enum\USBSTOR, VolumeInfoCache (time-correlated), setupapi
USB Connection Sessions	2015-03-24 13:58:33	SanDisk Cruzer Fit 4C530012550531106501 connected 2015-03-24 13:58:33.518909 -> unknown	DeviceClasses key last written (first connection after the last reboot)

5.2 Q2: Was the removable device reported in the DLP alert connected to the examined system(s)?

Conclusion: Not applicable - No removable media serial was supplied.

No specific findings were produced for this question; see the conclusion above and the coverage matrix.

5.3 Q3: Were the files of interest present on the examined system(s)? Were they deleted, renamed or archived?

Conclusion: **Not applicable** - No files of interest were supplied; see the document access findings.

No specific findings were produced for this question; see the conclusion above and the coverage matrix.

5.4 Q4: Is there evidence that the files were copied to, or opened from, removable media?

Conclusion: **Indicated** - Suspect workstation (INFORMANT-PC): 11 shortcut/jump-list records of files on removable media. Suspect workstation (INFORMANT-PC): 19 shellbag/MRU/Timeline records reference removable drives.

F-06 Files opened from removable media on Suspect workstation (INFORMANT-PC)

System: Suspect workstation (INFORMANT-PC)	Time: 2015-03-23 18:37:20 UTC	Supporting artifacts: 11	Removable media
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11 shortcut / jump list entries reference files on removable volumes. Windows creates these records when the user opens a file, so the files existed on the removable volume at the recorded times; the volume serial numbers tie them to the specific device.

WFA - File & Folder Access							
Shortcut / jump list entries pointing to removable media - Suspect workstation (INFORMANT-PC)							
Shortcut	Target path	Drive	Vol. serial	Label	First opened (UTC)	Last opened (UTC)	Source
[secret_project]_proposal.lnk	E:\RM#1\Secret Project Data...	FIXED	5C75-4D3E	Authorized...	2015-03-23 18:37:20	2015-03-23 18:37:20	LNK
[secret_project]_proposal.LNK	E:\RM#1\Secret Project Data...	FIXED	5C75-4D3E	Authorized...	2015-03-23 18:37:54	2015-03-23 18:37:54	LNK
[secret_project]_design_conc...	E:\RM#1\Secret Project Data...	FIXED	5C75-4D3E	Authorized...	2015-03-23 18:38:21	2015-03-23 18:38:21	LNK
[secret_project]_design_conc...	E:\RM#1\Secret Project Data...	FIXED	5C75-4D3E	Authorized...	2015-03-23 18:38:23	2015-03-23 18:38:23	LNK
47bb2136fda3f1ed.automatic...	E:\RM#1\Secret Project Data...	FIXED	5C75-4D3E	Authorized...	2015-03-23 18:37:54	2015-03-23 18:37:54	47bb2136fda3...
1b4dd67f29cb1962.automatic...	E:\RM#1\Secret Project Data...	FIXED	5C75-4D3E	Authorized...	2015-03-23 18:37:54	2015-03-23 18:37:54	Windows Expl...
4cc9bcff1a772a63.automatic...	E:\RM#1\Secret Project Data...	FIXED	5C75-4D3E	Authorized...	2015-03-23 18:38:21	2015-03-23 18:38:21	4cc9bcff1a772...
1b4dd67f29cb1962.automatic...	E:\	FIXED	5C75-4D3E	Authorized...	2015-03-23 18:38:21	2015-03-23 18:38:21	Windows Expl...
1b4dd67f29cb1962.automatic...	E:\RM#1\Secret Project Data...	FIXED	5C75-4D3E	Authorized...	2015-03-23 18:38:23	2015-03-23 18:38:23	Windows Expl...
1b4dd67f29cb1962.automatic...	E:\Secret Project Data\design...	FIXED	9E6A-5B82	IAMAN \$ _@	2015-03-24 14:01:23	2015-03-24 14:01:23	Windows Expl...
1b4dd67f29cb1962.automatic...	E:\SECRET~1\design\WINTe...				2015-03-24 14:01:29	2015-03-24 14:01:29	Windows Expl...
11 rows 0 highlighted							

1 Document opened from removable media

2 Volume serial of the removable volume

Figure 9: Shortcut / jump list entries pointing to removable media - Suspect workstation (INFORMANT-PC). LNK files are created when a file is opened; their link-info block records the drive type, volume serial number and label of the volume the file was on.

Supporting artifacts

Artifact	Time (UTC)	Summary	Source
Shortcut (LNK) Files	2015-03-23 18:37:20	LNK -> E:\RM#1\Secret Project Data\proposal\[secret_project]_proposal.docx (DRIVE_FIXED 5C75-4D3E)	C:\Users\informant\AppData\Roaming\Microsoft\Windows\Recent\[secret_project]_proposal.lnk
Shortcut (LNK) Files	2015-03-23 18:37:54	LNK -> E:\RM#1\Secret Project Data\proposal\[secret_project]_proposal.docx (DRIVE_FIXED 5C75-4D3E)	C:\Users\informant\AppData\Roaming\Microsoft\Office\Recent\[secret_project]_proposal.LNK
Shortcut (LNK) Files	2015-03-23 18:38:21	LNK -> E:\RM#1\Secret Project Data\design\[secret_project]_design_concept.ppt (DRIVE_FIXED 5C75-4D3E)	C:\Users\informant\AppData\Roaming\Microsoft\Windows\Recent\[secret_project]_design_concept.lnk
Shortcut (LNK) Files	2015-03-23 18:38:23	LNK -> E:\RM#1\Secret Project Data\design\[secret_project]_design_concept.ppt (DRIVE_FIXED 5C75-4D3E)	C:\Users\informant\AppData\Roaming\Microsoft\Office\Recent\[secret_project]_design_concept.LNK
Jump Lists	2015-03-23 18:37:54	Jump list [47bb2136fda3f1ed] E:\RM#1\Secret Project Data\proposal\[secret_project]_proposal.docx	C:\Users\informant\AppData\Roaming\Microsoft\Windows\Recent\AutomaticDestinations\47bb2136fda3f1ed.automaticDestinations

Artifact	Time (UTC)	Summary	Source
Jump Lists	2015-03-23 18:37:54	Jump list [Windows Explorer] E:\RM#1\Secret Project Data\proposal	C:\Users\informant\AppData\Roaming\Microsoft\Windows\Recent\AutomaticDestinations\1b4dd67f29cb1962.a
Jump Lists	2015-03-23 18:38:21	Jump list [4cc9bcff1a772a63] E:\RM#1\Secret Project Data\design\[secret_project]_design_concept.ppt	C:\Users\informant\AppData\Roaming\Microsoft\Windows\Recent\AutomaticDestinations\4cc9bcff1a772a63.a
Jump Lists	2015-03-23 18:38:21	Jump list [Windows Explorer] E:\	C:\Users\informant\AppData\Roaming\Microsoft\Windows\Recent\AutomaticDestinations\1b4dd67f29cb1962.a

F-07 Folder browsing and file use on removable drives - Suspect workstation (INFORMANT-PC)

System: Suspect workstation (INFORMANT-PC)	Time: -	Supporting artifacts: 19 Removable media
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19 shellbag, MRU, Timeline or prefetch records reference paths on removable volumes (E:), showing the user browsed and used content on them.

Artifact	Path / item	Time (UTC)	User
Shellbag (folder browsed)	E:\RM#1	2015-03-24 13:38:31	informant
Shellbag (folder browsed)	E:\RM#1\Secret Project Data	2015-03-24 13:38:52	informant
Shellbag (folder browsed)	E:\RM#1\Secret Project Data\design	2015-03-24 13:38:52	informant
Shellbag (folder browsed)	E:	2015-03-24 14:00:19	informant
Shellbag (folder browsed)	E:\Secret Project Data\technical review	2015-03-24 14:01:11	informant
Shellbag (folder browsed)	E:\Secret Project Data\proposal	2015-03-24 14:01:14	informant
Shellbag (folder browsed)	E:\Secret Project Data\progress	2015-03-24 14:01:15	informant
Shellbag (folder browsed)	E:\Secret Project Data\pricing decision	2015-03-24 14:01:17	informant
Shellbag (folder browsed)	E:\Secret Project Data\design	2015-03-24 14:01:29	informant
Shellbag (folder browsed)	E:\Secret Project Data\design\winter_whether_advisory.zip	2015-03-24 14:01:32	informant
Shellbag (folder browsed)	E:\Secret Project Data\design\winter_whether_advisory.zip\<UNKNOWN size=...	2015-03-24 14:01:32	informant
Shellbag (folder browsed)	E:\Secret Project Data	2015-03-24 20:54:07	informant
Office MRU	E:\RM#1\Secret Project Data\proposal\[secret_project]_proposal.docx	2015-03-23 18:37:54	informant
Office MRU	E:\RM#1\Secret Project Data\proposal\	2015-03-23 18:37:54	informant
Office MRU	E:\RM#1\Secret Project Data\design\[secret_project]_design_concept.ppt	2015-03-23 18:38:23	informant
Office MRU	E:\RM#1\Secret Project Data\design\	2015-03-23 18:38:23	informant
IE / Winlnet history (file ope...	E:\RM#1\Secret Project Data\proposal\[secret_project]_proposal.docx	2015-03-23 18:37:20	informant
IE / Winlnet history (file ope...	E:\RM#1\Secret Project Data\design\[secret_project]_design_concept.ppt	2015-03-23 18:38:21	informant
IE / Winlnet history (file ope...	E:\Secret Project Data\design\winter_whether_advisory.zip	2015-03-24 14:01:32	informant

19 rows | 0 highlighted

1 Folder / file on the removable drive

Figure 10: Other records referencing removable volumes - Suspect workstation (INFORMANT-PC). Shellbags record folders browsed in Explorer; MRU lists and the Timeline record files opened.

Supporting artifacts

Artifact	Time (UTC)	Summary	Source
Shellbags (folders browsed)	2015-03-24 13:38:31	Shellbag: E:\RM#1	UsrClass.dat BagMRU
Shellbags (folders browsed)	2015-03-24 13:38:52	Shellbag: E:\RM#1\Secret Project Data	UsrClass.dat BagMRU
Shellbags (folders browsed)	2015-03-24 13:38:52	Shellbag: E:\RM#1\Secret Project Data\design	UsrClass.dat BagMRU
Shellbags (folders browsed)	2015-03-24 14:00:19	Shellbag: E:	UsrClass.dat BagMRU

Artifact	Time (UTC)	Summary	Source
Shellbags (folders browsed)	2015-03-24 14:01:11	Shellbag: E:\Secret Project Data\technical review	UsrClass.dat BagMRU
Shellbags (folders browsed)	2015-03-24 14:01:14	Shellbag: E:\Secret Project Data\proposal	UsrClass.dat BagMRU
Shellbags (folders browsed)	2015-03-24 14:01:15	Shellbag: E:\Secret Project Data\progress	UsrClass.dat BagMRU
Shellbags (folders browsed)	2015-03-24 14:01:17	Shellbag: E:\Secret Project Data\pricing decision	UsrClass.dat BagMRU

5.5 Q5: Is there evidence that the files were uploaded via webmail, cloud storage or file-sharing websites?

Conclusion: Indicated - Suspect workstation (INFORMANT-PC): browser file dialogs were used.

F-08 Browser file dialog activity - Suspect workstation (INFORMANT-PC)

System: Suspect workstation (INFORMANT-PC)	Time: 2015-03-25 15:28:33 UTC	Supporting artifacts: 1 Web activity
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LastVisitedPidIMRU shows a web browser opened a file dialog (used for uploading attachments or saving downloads) in the listed folder(s). Where OpenSavePidIMRU recorded a file selection at the same moment, the selected file is shown.

Application	Last folder used	Key last written (UTC)	Files of interest in folder	File picked (OpenSave MRU)
iexplore.exe	Download\	2015-03-25 15:28:33	1E11-Windows6.1-x64-en-us.exe	

Figure 11: Browser file-dialog usage - Suspect workstation (INFORMANT-PC)

Registry - ComDlg32 MRU entries (decoded)

Key (HKCU\...\Explorer\ComDlg32)	MRU position	Decoded entry	Key last written (UTC)
LastVisitedPidIMRU	1	iexplore.exe Download\	2015-03-25 15:28:33
OpenSavePidIMRU*	4	Download\IE11-Windows6.1-x64-en-us.exe	2015-03-25 15:28:33
OpenSavePidIMRU*	2	Download\Eraser 6.2.0.2962.exe	2015-03-25 15:28:33
OpenSavePidIMRU*	1	Download\ccsetup504.exe	2015-03-25 15:28:33

Figure 12: Registry - ComDlg32 MRU entries (decoded). LastVisitedPidIMRU records the last folder each application used in an Open/Save dialog; OpenSavePidIMRU records the files selected, per extension. MRU position 0 is the most recent entry, written at the key's last-written time.

Supporting artifacts

Artifact	Time (UTC)	Summary	Source
Last Visited Folder MRU (by application)		iexplore.exe last used folder Download\	C:\Users\informant\ntuser.dat\Software\Microsoft\Windows\CurrentVersion\Explorer\ComDlg32\LastVisitedPidIMRU

F-10 Files synchronized to cloud storage from Suspect workstation (INFORMANT-PC)

System: Suspect workstation (INFORMANT-PC)	Time: 2015-03-23 20:05:37 UTC	Supporting artifacts: 8 Cloud
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The Google Drive (legacy) client recorded 8 completed sync operations; 2 upload(s) of local files to the cloud account: do_u_wanna_build_a_snow_man.mp3, happy_holiday.jpg. Later deletions or sharing changes of the same files are listed with them.

Windows Forensic Automation - Artifact Viewer

Cloud client sync log - completed operations

Filter rows...

Time (UTC)	Service	Operation	File	Local folder	Account
2015-03-23 20:05:37	Google Drive (legacy)	downloaded from Google Dr...	How to get started with Drive		iaman.informant.pers...
2015-03-23 20:06:08	Google Drive (legacy)	deleted from Google Drive (l...	How to get started with Drive	C:\Users\informant\Google ...	iaman.informant.pers...
2015-03-23 20:32:...	Google Drive (legacy)	uploaded to Google Drive	happy_holiday.jpg	C:\Users\informant\Google ...	iaman.informant.pers...
2015-03-23 20:32:...	Google Drive (legacy)	uploaded to Google Drive	do_u_wanna_build_a_snow_...	C:\Users\informant\Google ...	iaman.informant.pers...
2015-03-23 20:34:25	Google Drive (legacy)	sharing settings changed in ...	do_u_wanna_build_a_snow_...		iaman.informant.pers...
2015-03-23 20:34:35	Google Drive (legacy)	sharing settings changed in ...	happy_holiday.jpg		iaman.informant.pers...
2015-03-23 20:42:20	Google Drive (legacy)	deleted from Google Drive (l...	happy_holiday.jpg	C:\Users\informant\Google ...	iaman.informant.pers...
2015-03-23 20:42:22	Google Drive (legacy)	deleted from Google Drive (l...	do_u_wanna_build_a_snow_...	C:\Users\informant\Google ...	iaman.informant.pers...

8 rows | 2 highlighted

1 Local file uploaded to the cloud account

Figure 13: Cloud client sync log - completed operations

Supporting artifacts

Artifact	Time (UTC)	Summary	Source
Cloud Synced Items	2015-03-23 20:05:37	Google Drive downloaded from Google Drive: How to get started with Drive	C:\Users\informant\AppData\Local\Google\Drive\user_default\sync_log.log
Cloud Synced Items	2015-03-23 20:06:08	Google Drive deleted from Google Drive (local deletion synchronized): How to get started with Drive	C:\Users\informant\AppData\Local\Google\Drive\user_default\sync_log.log
Cloud Synced Items	2015-03-23 20:32:45	Google Drive uploaded to Google Drive: happy_holiday.jpg	C:\Users\informant\AppData\Local\Google\Drive\user_default\sync_log.log
Cloud Synced Items	2015-03-23 20:32:45	Google Drive uploaded to Google Drive: do_u_wanna_build_a_snow_man.mp3	C:\Users\informant\AppData\Local\Google\Drive\user_default\sync_log.log
Cloud Synced Items	2015-03-23 20:34:25	Google Drive sharing settings changed in the cloud: do_u_wanna_build_a_snow_man.mp3	C:\Users\informant\AppData\Local\Google\Drive\user_default\sync_log.log
Cloud Synced Items	2015-03-23 20:34:35	Google Drive sharing settings changed in the cloud: happy_holiday.jpg	C:\Users\informant\AppData\Local\Google\Drive\user_default\sync_log.log
Cloud Synced Items	2015-03-23 20:42:20	Google Drive deleted from Google Drive (local deletion synchronized): happy_holiday.jpg	C:\Users\informant\AppData\Local\Google\Drive\user_default\sync_log.log
Cloud Synced Items	2015-03-23 20:42:22	Google Drive deleted from Google Drive (local deletion synchronized): do_u_wanna_build_a_snow_man.mp3	C:\Users\informant\AppData\Local\Google\Drive\user_default\sync_log.log

MITRE ATT&CK: T1567.002

F-15 Web session / file traces in pagefile or hibernation file - Suspect workstation (INFORMANT-PC)

System: Suspect workstation (INFORMANT-PC)	Time: -	Supporting artifacts: 13 Memory artifacts
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13 relevant strings were recovered from memory-backed files (pagefile.sys / hiberfil.sys / swapfile.sys), which persist after browser history is cleared: webmail and attachment URLs, cloud destinations and paths of files of interest.

File	Kind	Value	Service / reason	Count	Offset
hiberfil.sys	URL	https://drive.google.com	Google Drive	1	0x5fc65b50
hiberfil.sys	URL	https://docs.google.com	Google Drive	1	0x5fc65b6c
hiberfil.sys	E-mail addr...	ipc.invalidation@gmail.com	webmail provider address	1	0x5fcea468
hiberfil.sys	URL	https://docs.google.com/feeds/	Google Drive	1	0x5fd9978c
hiberfil.sys	URL	https://docs.google.com/	Google Drive	1	0x5fdb6a3c
pagefile.sys	E-mail addr...	actualspyrep@gmail.com	webmail provider address	1	0x9a40b0
pagefile.sys	E-mail addr...	bugzreportz@gmail.com	webmail provider address	1	0x11151e0
pagefile.sys	E-mail addr...	hola@hotmail.com	webmail provider address	1	0x112ee20
pagefile.sys	E-mail addr...	post0001@gmail.com	webmail provider address	1	0x165a238
pagefile.sys	E-mail addr...	%s@hotmail.com	webmail provider address	1	0x170198c
pagefile.sys	File path	E:\RM#1\Secret Project Data\proposal\[secret_project]_proposal.docx	case keyword	1	0x61358d4
pagefile.sys	File path	E:\RM#1\Secret Project Data\design\[secret_project]_design_concept.ppt	case keyword	1	0x61385f0
pagefile.sys	URL	https://outlook.office365.com/OAB/818caab3-a88f-48e9-abb5-a161d4c7043a/...	Outlook on the web (M3...	2	0xaf36d5c

13 rows | 0 highlighted

Figure 14: Strings recovered from memory files

Supporting artifacts

Artifact	Time (UTC)	Summary	Source
Memory File Strings (pagefile / hiberfil)		hiberfil.sys: URL https://drive.google.com	C:\hiberfil.sys @ 0x5fc65b50
Memory File Strings (pagefile / hiberfil)		hiberfil.sys: URL https://docs.google.com	C:\hiberfil.sys @ 0x5fc65b6c
Memory File Strings (pagefile / hiberfil)		hiberfil.sys: E-mail address ipc.invalidation@gmail.com	C:\hiberfil.sys @ 0x5fcea468
Memory File Strings (pagefile / hiberfil)		hiberfil.sys: URL https://docs.google.com/feeds/	C:\hiberfil.sys @ 0x5fd9978c
Memory File Strings (pagefile / hiberfil)		hiberfil.sys: URL https://docs.google.com/	C:\hiberfil.sys @ 0x5fdb6a3c
Memory File Strings (pagefile / hiberfil)		pagefile.sys: E-mail address actualspyrep@gmail.com	C:\pagefile.sys @ 0x9a40b0
Memory File Strings (pagefile / hiberfil)		pagefile.sys: E-mail address bugzreportz@gmail.com	C:\pagefile.sys @ 0x11151e0
Memory File Strings (pagefile / hiberfil)		pagefile.sys: E-mail address hola@hotmail.com	C:\pagefile.sys @ 0x112ee20

5.6 Q6: Were other channels used (e-mail, cloud sync client, CD/DVD burning, printing, archiving / transfer tools)?

Conclusion: Indicated - Suspect workstation (INFORMANT-PC): e-mail correspondence with spy.conspirator@nist.gov (15), iaman.informant@nist.gov (14). Suspect workstation (INFORMANT-PC): 2 file upload(s) by the Google Drive (legacy) client (do_u_wanna_build_a_snow_man.mp3, happy_holiday.jpg). Suspect workstation (INFORMANT-PC): The cdrom driver logged 4 completed disc write(s) (System event 133) between 2015-03-24 19:47:47 and 2015-03-24 20:41:21 UTC. \$UsnJrnl records 8 disc-mastering session(s) (IMAPI DAT/FIL/POST temporary files). 62 file(s) were placed in the Explorer burn staging folder (AppData\Local\Microsoft\Windows\Burn\Burn). Suspect workstation (INFORMANT-PC): Google Drive / Backup and Sync (legacy client) executed.

F-09 E-mail correspondence on Suspect workstation (INFORMANT-PC)

System: Suspect workstation (INFORMANT-PC)	Time: 2015-03-23 18:44:00 UTC	Supporting artifacts: 9 E-mail
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The mail store holds 14 messages. Listed below are 9 sent, deleted or attachment-carrying messages, including 3 message(s) that the Windows Search index recorded but that are no longer in the mailbox (deleted). Addresses seen: spy.conspirator@nist.gov (15), iaman.informant@nist.gov (14).

E-mail messages - sent, deleted or with attachments

Time (UTC)	Folder	From	To	Subject	Attachments	Source
2015-03-23 18:44:00	Sent Items	iaman (Exchange mailbox)	spy	RE: Hello, iaman		Mailbox
2015-03-23 19:27:00	Sent Items	iaman (Exchange mailbox)	spy	RE: Important request		Mailbox
2015-03-23 20:38:47	Sent Items (not in mailb...	(mailbox owner)	spy.conspirator@nist.gov	It's me		Windows Search index
2015-03-23 20:41:22	Deleted Items	spy <spy <spy.conspirato...	iaman <iaman.informant...	RE: It's me		Mailbox
2015-03-24 13:30:10	Sent Items (not in mailb...	(mailbox owner)	spy.conspirator@nist.gov	RE: Last request		Windows Search index
2015-03-24 13:35:00	Deleted Items	iaman (Exchange mailbox)	spy	RE: Last request		Mailbox
2015-03-24 19:32:42	Inbox (not in mailbox)	16DEB821A6D2423FB72...	iaman.informant@nist.g...	Watch out!		Windows Search index
2015-03-24 19:34:00	Deleted Items	iaman (Exchange mailbox)	spy	RE: Watch out!		Mailbox
2015-03-24 21:05:00	Deleted Items	iaman (Exchange mailbox)	spy	Done		Mailbox

1 Message deleted

Figure 15: E-mail messages - sent, deleted or with attachments

Supporting artifacts

Artifact	Time (UTC)	Summary	Source
E-mail Messages	2015-03-23 18:44:00	[/Root - Mailbox/IPM_SUBTREE/Sent Items] RE: Hello, iaman from iaman (Exchange mailbox) to spy	C:\Users\informant\AppData\Local\Microsoft\Outlook\iaman.informant@nist.gov.ost
E-mail Messages	2015-03-23 19:27:00	[/Root - Mailbox/IPM_SUBTREE/Sent Items] RE: Important request from iaman (Exchange mailbox) to spy	C:\Users\informant\AppData\Local\Microsoft\Outlook\iaman.informant@nist.gov.ost
E-mail Messages	2015-03-23 20:41:22	[/Root - Mailbox/IPM_SUBTREE/Deleted Items] RE: It's me from spy <spy <spy.conspirator@nist.gov>> to iaman <iaman.informant@nist.gov>	C:\Users\informant\AppData\Local\Microsoft\Outlook\iaman.informant@nist.gov.ost
E-mail Messages	2015-03-24 13:35:00	[/Root - Mailbox/IPM_SUBTREE/Deleted Items] RE: Last request from iaman (Exchange mailbox) to spy	C:\Users\informant\AppData\Local\Microsoft\Outlook\iaman.informant@nist.gov.ost
E-mail Messages	2015-03-24 19:34:00	[/Root - Mailbox/IPM_SUBTREE/Deleted Items] RE: Watch out! from iaman (Exchange mailbox) to spy	C:\Users\informant\AppData\Local\Microsoft\Outlook\iaman.informant@nist.gov.ost
E-mail Messages	2015-03-24 21:05:00	[/Root - Mailbox/IPM_SUBTREE/Deleted Items] Done from iaman (Exchange mailbox) to spy	C:\Users\informant\AppData\Local\Microsoft\Outlook\iaman.informant@nist.gov.ost
Windows Search Index Items	2015-03-23 20:38:51	Indexed e-mail 'It's me' from to spy.conspirator@nist.gov	C:\ProgramData\Microsoft\Search\Data\Applications\Windows\Windows.edb
Windows Search Index Items	2015-03-24 13:30:14	Indexed e-mail 'RE: Last request' from to spy.conspirator@nist.gov	C:\ProgramData\Microsoft\Search\Data\Applications\Windows\Windows.edb

F-11 CD / DVD burning and optical media on Suspect workstation (INFORMANT-PC)

System: Suspect workstation (INFORMANT-PC) Time: 2015-03-24 19:46:16 UTC Supporting artifacts: 84 | Optical media

The cdrom driver logged 4 completed disc write(s) (System event 133) between 2015-03-24 19:47:47 and 2015-03-24 20:41:21 UTC. \$UsnJrnl records 8 disc-mastering session(s) (IMAPI DAT/FIL/POST temporary files). 62 file(s) were placed in the Explorer burn staging folder (AppData\Local\Microsoft\Windows\Burn\Burn). The burn method last selected was 0 (Live File System - like a USB flash drive) (registry key written 2015-03-24 20:53:16 UTC). Shellbags show the Explorer disc-burning view was opened (1 record(s)). 8 shortcut / jump-list record(s) point to files on optical media (volume IAMAN CD VSN A58AD72A, IAMAN CD VSN AE88F176, IAMAN CD VSN E297F1A5).

Windows Forensic Automation - Artifact Viewer

Disc writing activity and files opened from optical media

Filter rows...

Time (UTC)	Source	Event	Item	User
2015-03-24 19:46:16	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\...	informant
2015-03-24 19:46:19	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\...	informant
2015-03-24 19:46:23	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\...	informant
2015-03-24 19:46:23	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\...	informant
2015-03-24 19:46:23	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\...	informant
2015-03-24 19:46:25	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\...	informant
2015-03-24 19:46:26	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\...	informant
2015-03-24 19:46:26	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\...	informant
2015-03-24 19:46:27	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\...	informant
2015-03-24 19:46:27	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\...	informant
2015-03-24 19:46:29	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\...	informant
2015-03-24 19:46:29	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\...	informant
2015-03-24 19:46:29	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\...	informant
2015-03-24 19:46:29	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\...	informant
2015-03-24 19:46:29	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\...	informant
2015-03-24 19:46:29	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\...	informant
2015-03-24 19:46:29	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\...	informant
2015-03-24 19:46:29	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\...	informant
2015-03-24 19:46:29	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\...	informant
2015-03-24 19:47:47	\$UsnJrnl	Disc mastering temporary files	DAT34216.tmp, DAT67383.tmp, FIL39751.tmp, POST39751.tmp	
2015-03-24 19:47:...	System event log (cdrom 133)	Disc write completed	\Device\CdRom0	
2015-03-24 19:49:26	\$UsnJrnl	Disc mastering temporary files	DAT34216.tmp, FIL39751.tmp, POST39751.tmp	
2015-03-24 19:54:35	Shellbag	Explorer disc-burning view open...	D:\<CDBURN>	informant
2015-03-24 19:55:15	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\...	informant

40 rows (showing first 22) | 4 highlighted

1 Disc write recorded by the cdrom driver

Figure 16: Disc writing activity and files opened from optical media

Supporting artifacts

Artifact	Time (UTC)	Summary	Source
Files Staged for CD/DVD Burning	2015-03-24 19:46:16	Staged for burning: C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\design\winter_storm.amr	\$UsnJrnl:\$J
Files Staged for CD/DVD Burning	2015-03-24 19:46:19	Staged for burning: C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\design\winter_whether_advisory.zip	\$UsnJrnl:\$J
Files Staged for CD/DVD Burning	2015-03-24 19:46:23	Staged for burning: C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\pricing decision\my_favorite_cars.db	\$UsnJrnl:\$J
Files Staged for CD/DVD Burning	2015-03-24 19:46:23	Staged for burning: C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\pricing decision\my_favorite_movies.7z	\$UsnJrnl:\$J
Files Staged for CD/DVD Burning	2015-03-24 19:46:23	Staged for burning: C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\pricing decision\new_years_day.jpg	\$UsnJrnl:\$J
Files Staged for CD/DVD Burning	2015-03-24 19:46:25	Staged for burning: C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\pricing decision\super_bowl.avi	\$UsnJrnl:\$J
Files Staged for CD/DVD Burning	2015-03-24 19:46:26	Staged for burning: C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\progress\my_friends.svg	\$UsnJrnl:\$J
Files Staged for CD/DVD Burning	2015-03-24 19:46:26	Staged for burning: C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\progress\my_smartphone.png	\$UsnJrnl:\$J

MITRE ATT&CK: T1052

F-12 Archiving / data transfer tools used on Suspect workstation (INFORMANT-PC)

System: Suspect workstation (INFORMANT-PC)	Time: -	Supporting artifacts: 2 Staging
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Execution evidence exists for Google Drive / Backup and Sync (legacy client).

Tool	Artifact	Path	Time (UTC)	User
Google Drive / Back...	userassist	C:\Program Files (x86)\Google\Drive\googledrivesync.exe	2015-03-25 15:21:30	informant
Google Drive / Back...	prefetch	GOOGLEDRIVESYNC.EXE	2015-03-25 15:21:31	

2 rows | 2 highlighted

Figure 17: Archiver / transfer tool execution

Supporting artifacts

Artifact	Time (UTC)	Summary	Source
Prefetch	2015-03-25 15:21:31	GOOGLEDRIVESYNC.EXE ran 2x, last 2015-03-25 15:21:31.899405	C:\Windows\Prefetch\GOOGLEDRIVE SYNC.EXE-841A0D94.pf
UserAssist (GUI program execution)	2015-03-25 15:21:30	UserAssist: C:\Program Files (x86)\Google\Drive\googledrivesync.exe (runs 1)	NTUSER\...\UserAssist\{CEBFF5CD-ACE2-4F4F-9178-9926F41749EA}\Count

F-14 Cloud storage clients configured on Suspect workstation (INFORMANT-PC)

System: Suspect workstation (INFORMANT-PC)	Time: -	Supporting artifacts: 1 Cloud
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1 cloud storage account(s) / clients were configured.

Service	Account	Local sync folder	Config time (UTC)
Google Drive (legacy client)	iaman.informant.personal@gmail.com		2015-03-25 15:22:48

1 rows | 0 highlighted

Figure 18: Cloud storage accounts

Supporting artifacts

Artifact	Time (UTC)	Summary	Source
Cloud Storage Accounts	2015-03-25 15:22:48	Google Drive (legacy client) account iaman.informant.personal@gmail.com - sync folder (not recorded)	C:\Users\informant\AppData\Local\Google\Drive\user_default\sync_config.db

5.7 Q7: Did the files reach the personal / secondary device, and how?

Conclusion: Not applicable - No personal / secondary device was examined.

No specific findings were produced for this question; see the conclusion above and the coverage matrix.

5.8 Q8: Do the system artifacts corroborate each DLP alert?

Conclusion: Not applicable - No DLP alert export was supplied.

No specific findings were produced for this question; see the conclusion above and the coverage matrix.

5.9 Q9: Is there evidence of attempts to conceal the activity or remove traces?

Conclusion: Yes - Suspect workstation (INFORMANT-PC): CCleaner; Eraser; Search / page about hiding traces; Secure deletion pattern.

F-03 Anti-forensic activity indicators on Suspect workstation (INFORMANT-PC)

System: Suspect workstation (INFORMANT-PC)

Time: 2015-03-23 18:16:55 UTC

Supporting artifacts: 150 | Anti-forensics

180 indicators of attempts to remove or obscure traces were identified (23 strong): CCleaner; Eraser; File deleted via Recycle Bin; Search / page about hiding traces; Secure deletion pattern; System time changed; Timestamp anomaly.

Anti-forensics indicators - Suspect workstation (INFORMANT-PC)			
Time (UTC)	Indicator	Detail	Strength
2015-03-23 18:16:...	Search / page about hiding traces (web search)	how to delete data	strong
2015-03-23 18:16:...	Search / page about hiding traces (web visit)	how to delete data - Google Search	strong
2015-03-23 18:17:...	Search / page about hiding traces (web search)	anti-forensics	strong
2015-03-23 18:17:...	Search / page about hiding traces (web visit)	anti-forensics - Google Search	strong
2015-03-23 18:17:...	Search / page about hiding traces (web visit)		strong
2015-03-23 18:17:...	Search / page about hiding traces (web visit)	Anti-forensic techniques - ForensicsWiki	strong
2015-03-23 18:17:...	Search / page about hiding traces (web visit)		strong
2015-03-23 18:18:...	Search / page about hiding traces (web visit)		strong
2015-03-25 14:46:...	Search / page about hiding traces (web visit)		strong
2015-03-25 14:47:...	Search / page about hiding traces (web visit)		strong
2015-03-25 14:47:...	Search / page about hiding traces (web visit)		strong
2015-03-25 14:48:...	Search / page about hiding traces (web visit)		strong
2015-03-25 14:48:...	Search / page about hiding traces (web visit)		strong
2015-03-25 14:48:...	Search / page about hiding traces (web visit)		strong
2015-03-25 14:48:...	Search / page about hiding traces (web visit)		strong
2015-03-25 14:48:...	Search / page about hiding traces (web visit)		strong
2015-03-25 14:48:...	Search / page about hiding traces (web visit)		strong
2015-03-25 14:48:...	Search / page about hiding traces (web visit)		strong
2015-03-25 14:57:...	Eraser (installed_program)	Eraser 6.2.0.2962	strong
2015-03-25 15:12:...	Eraser (userassist)	C:\Program Files\Eraser\Eraser.exe	strong
2015-03-25 15:13:...	Eraser (prefetch)	ERASER.EXE	strong
2015-03-25 15:13:...	Secure deletion pattern (\$UsnJrnl: renamed to rando...	C:\Users\informant\Desktop\temp - renamed 7 times (t6uk, 2FDM, hA_...	strong
2015-03-25 15:15:...	CCleaner (prefetch)	CCLEANER64.EXE	strong

1 Attempt to remove traces

Figure 19: Anti-forensics indicators - Suspect workstation (INFORMANT-PC)

Supporting artifacts

Artifact	Time (UTC)	Summary	Source
Prefetch	2015-03-25 15:13:30	ERASER.EXE ran 2x, last 2015-03-25 15:13:30.777004	C:\Windows\Prefetch\ERASER.EXE-CE61944A.pf
Prefetch	2015-03-25 15:15:50	CCLEANER64.EXE ran 2x, last 2015-03-25 15:15:50.711605	C:\Windows\Prefetch\CCLEANER64.EXE-779BD542.pf
UserAssist (GUI program execution)	2015-03-25 15:12:28	UserAssist: C:\Program Files\Eraser\Eraser.exe (runs 1)	NTUSER\...\UserAssist\{CEBFF5CD-ACE2-4F4F-9178-9926F41749EA}\Count
UserAssist (GUI program execution)	2015-03-25 15:15:50	UserAssist: C:\Program Files\CCleaner\CCleaner64.exe (runs 1)	NTUSER\...\UserAssist\{CEBFF5CD-ACE2-4F4F-9178-9926F41749EA}\Count
Installed Programs	2015-03-25 14:57:31	Installed: Eraser 6.2.0.2962	HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Uninstall\{C6E287F1-2E47-45F0-BB51-94F815CFFB48}
Shimcache (AppCompatCache)	2015-01-12 22:56:36	Shimcache #19: \??\C:\Program Files\Eraser\Eraser.exe	SYSTEM\ControlSet001\Control\Session Manager\AppCompatCache

Artifact	Time (UTC)	Summary	Source
Shimcache (AppCompatCache)	2015-03-13 11:10:26	Shimcache #61: \\?\C:\Program Files\CCleaner\CCleaner64.exe	SYSTEM\ControlSet001\Control\Session Manager\AppCompatCache
Shimcache (AppCompatCache)	2015-03-13 11:10:26	Shimcache #62: \\?\C:\Program Files\CCleaner\CCleaner.exe	SYSTEM\ControlSet001\Control\Session Manager\AppCompatCache

MITRE ATT&CK: T1070

5.10 Q10: Do residual data areas (unallocated space, slack, pagefile, hibernation file) contain traces of the files or activity?

Conclusion: Yes - Suspect workstation (INFORMANT-PC): 13 relevant strings in memory files. Suspect workstation (INFORMANT-PC): hits in \$I30 directory index (incl. index slack), \$LogFile (NTFS transaction log), \$UsnJrnl (change journal), File slack, MFT record (resident data / metadata), Unallocated (deleted file remnant), Unallocated space, Volume Shadow Copy store, pagefile.sys.

F-13 Keyword search results - Suspect workstation (INFORMANT-PC)

System: Suspect workstation (INFORMANT-PC)	Time: -	Supporting artifacts: 0 Keyword search
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3 keywords (examiner supplied and derived from the files of interest, device serials and destinations) were searched across every byte of the image. 1,152 physical hits were recorded; 529 of them are in residual areas (\$I30 directory index (incl. index slack), \$LogFile (NTFS transaction log), \$UsnJrnl (change journal), File slack, MFT record (resident data / metadata), Unallocated (deleted file remnant), Unallocated space, Volume Shadow Copy store, pagefile.sys), i.e. traces that survive deletion or are invisible to the file system.

Keyword search summary - Suspect workstation (INFORMANT-PC)

Keyword	Origin	Hits	In residual areas	Hits by area
secret_project	keyword	525	406	\$UsnJrnl (change journal) [C:]: 210; Allocated file [C:]: 119; Volume Shadow Copy store [...]
spy.conspirator	keyword	5	1	Allocated file [C:]: 4; File slack [C:]: 1
confidential	keyword	2234	122	Allocated file [C:]: 2112; Unallocated space [C:]: 59; File slack [C:]: 36; pagefile.sys [C:]: 1...

Figure 20: Keyword search summary - Suspect workstation (INFORMANT-PC). Physical search of every byte of the image (ASCII/UTF-8 and UTF-16LE) plus document text

Windows Forensic Automation - Hex View [File slack]

Evidence: Suspect workstation (INFORMANT-PC)
Volume: C:
Area: File slack

Volume offset: 1,476,728,358 (0x58051626)
Encoding: utf16le
Attributed to: C:\Users\informant\AppData\Roaming\Microsoft...

Offset	00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F	Decoded text
0000580515E0																	
0000580515F0																	
000058051600	65	00	6E	00	74	00	5F	00	30	00	30	00	30	00	30	00	e.n.t._.0.0.0.
000058051610	35	00	5F	00	30	00	30	00	39	00	5F	00	4E	00	41	00	5._.0.9..N.A.
000058051620	4D	00	45	00	3D	00	43	00	6F	00	6E	00	66	00	69	00	M.E.=C.o.n.f.i.
000058051630	64	00	65	00	6E	00	74	00	69	00	61	00	6C	00	42	00	d.e.n.t.i.a.l.B.
000058051640	79	00	74	00	65	00	73	00	53	00	65	00	6E	00	74	00	y.t.e.s.s.e.n.t.
000058051650	0D	00	0A	00	43	00	6F	00	6E	00	66	00	69	00	64	00	...C.o.n.f.i.d.
000058051660	65	00	6E	00	74	00	69	00	61	00	6C	00	42	00	79	00	e.n.t.i.a.l.B.y.
000058051670	74	00	65	00	73	00	53	00	65	00	6E	00	74	00			t.e.s.s.e.n.t.

Selection: 24 bytes at 1,476,728,358 (0x58051626) | Encoding: utf16le

1 Keyword hit (utf16le)

Figure 21: 'confidential' in File slack

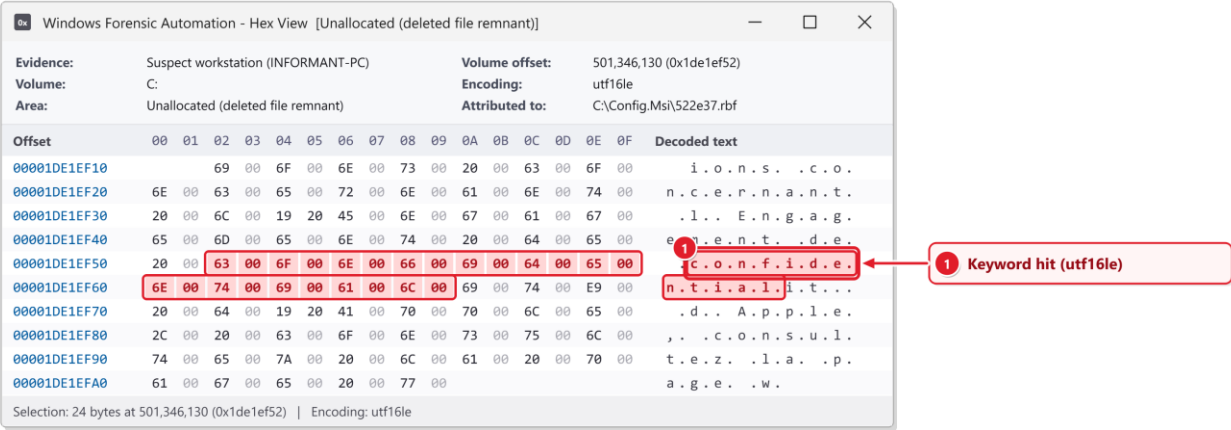


Figure 22: 'confidential' in Unallocated (deleted file remnant)

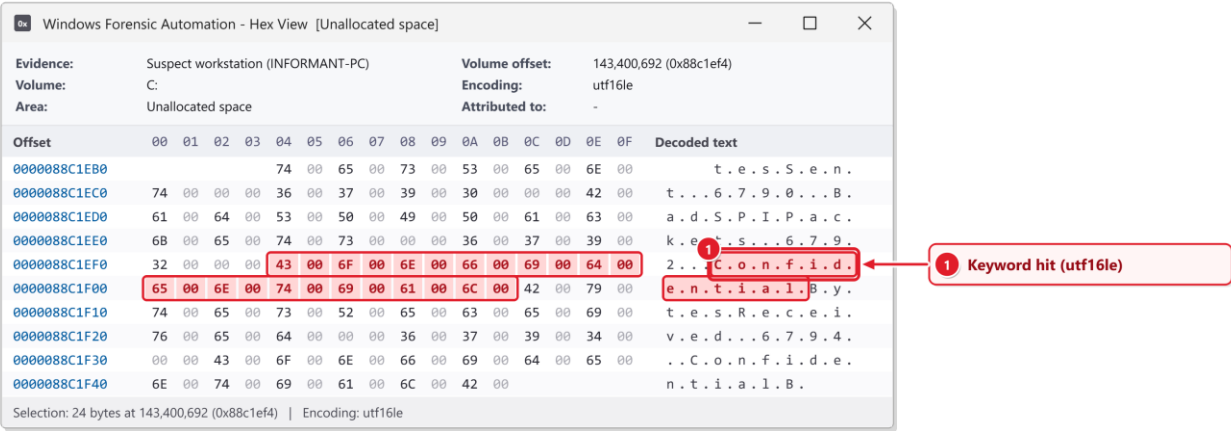


Figure 23: 'confidential' in Unallocated space

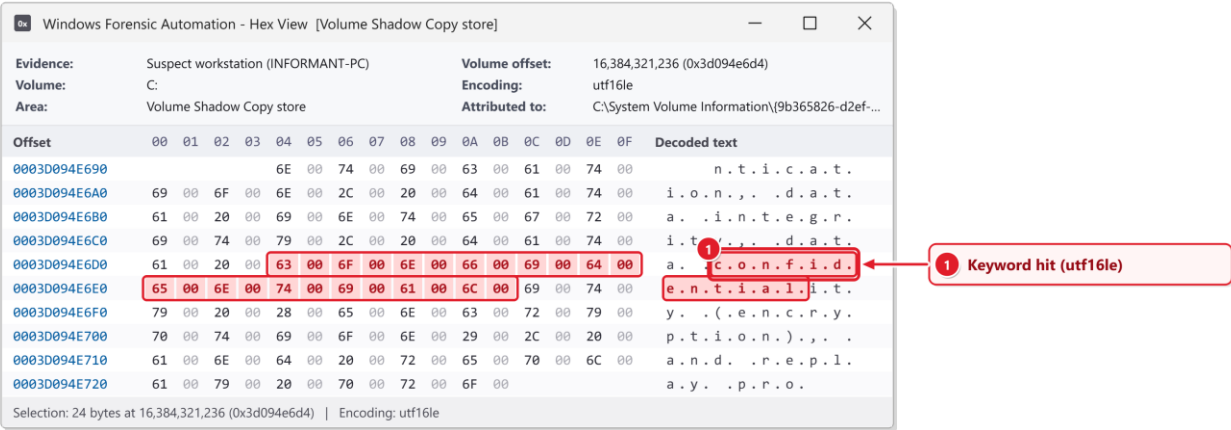


Figure 24: 'confidential' in Volume Shadow Copy store

5.11 Other observations

F-01 System overview - Suspect workstation (INFORMANT-PC)

System: Suspect workstation (INFORMANT-PC)	Time: -	Supporting artifacts: 6 System
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Suspect workstation (INFORMANT-PC) runs Windows 7 Ultimate (build 7601), installed 2015-03-22 14:34:26 UTC (2015-03-22 10:34:26 UTC-04:00), configured for time zone Eastern Standard Time. User profiles: informant, admin11, temporary. 6 local accounts and 66 installed programs were recorded.

The screenshot shows a window titled 'WFA - System Information' with a sub-header 'Suspect workstation (INFORMANT-PC) - system details'. It contains a table with system properties and their values.

Property	Value
Computer name	INFORMANT-PC
Operating system	Windows 7 Ultimate (build 7601)
Installed (UTC)	2015-03-22 14:34:26
Registered owner	informant
Time zone	Eastern Standard Time (America/New_York)
Last shutdown (UTC)	2015-03-25 15:31:05
IP address(es)	10.11.11.129
Domain	-
Last logged on user	.\informant
ClearPageFileAtShutdown	0
HibernateEnabled	1

11 rows | 0 highlighted

Figure 25: Suspect workstation (INFORMANT-PC) - system details

Local user accounts (SAM)

Account	RID	Created	Last logon	Logons	Password set	Flags
Guest	501	2015-03-25 10:33:22		0		Disabled, Password not require...
ITechTeam	1002	2015-03-22 15:52:30		0	2015-03-22 15:52:45	Normal account, Password doe...
Administrator	500	2015-03-25 10:33:22	2010-11-21 03:47:20	6	2010-11-21 03:57:24	Disabled, Normal account, Pass...
temporary	1003	2015-03-22 15:53:01	2015-03-22 15:55:57	1	2015-03-22 15:53:11	Normal account, Password doe...
admin11	1001	2015-03-22 15:51:54	2015-03-22 15:57:02	2	2015-03-22 15:52:10	Normal account, Password doe...
informant	1000	2015-03-22 14:33:54	2015-03-25 14:45:59	10	2015-03-22 14:33:54	Password not required, Normal...

Figure 26: Local user accounts (SAM)

Supporting artifacts

Artifact	Time (UTC)	Summary	Source
Local User Accounts		Account Guest (RID 501) logons=0	SAM\Domains\Account\Users\000001F5
Local User Accounts		Account ITechTeam (RID 1002) logons=0	SAM\Domains\Account\Users\000003EA
Local User Accounts	2010-11-21 03:47:20	Account Administrator (RID 500) logons=6	SAM\Domains\Account\Users\000001F4
Local User Accounts	2015-03-22 15:55:57	Account temporary (RID 1003) logons=1	SAM\Domains\Account\Users\000003EB
Local User Accounts	2015-03-22 15:57:02	Account admin11 (RID 1001) logons=2	SAM\Domains\Account\Users\000003E9
Local User Accounts	2015-03-25 14:45:59	Account informant (RID 1000) logons=10	SAM\Domains\Account\Users\000003E8

F-02 Network share access on Suspect workstation (INFORMANT-PC)

System: Suspect workstation (INFORMANT-PC)	Time: 2015-03-23 20:26:04 UTC	Supporting artifacts: 22 Network shares
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22 records show network shares being mapped or browsed, or files on them being opened; shares:

\\10.11.11.128\SECURED_DRIVE, \\10.11.11.128\secured_drive; drive letters mapped to shares: V:. Files opened from a share and later seen on removable media or in a sync folder identify where the data was taken from.

Windows Forensic Automation - Artifact Viewer

Network share activity - Suspect workstation (INFORMANT-PC)

Filter rows...

Artifact	Share / path	Time (UTC)	User
Mapped network drive	\\10.11.11.128\secured_drive	2015-03-23 20:26:04	informant
MountPoints2 (share opened by ...)	\\10.11.11.128\secured_drive	2015-03-23 20:26:04	informant
IE / WinInet history (file opened)	\\10.11.11.128\secured_drive\Secret Project Data\pricing decision\secret_proj...	2015-03-23 20:26:53	informant
Shortcut (LNK)	\\10.11.11.128\SECURED_DRIVE\Secret Project Data\pricing decision\secret_p...	2015-03-23 20:26:53	informant
Jump list	\\10.11.11.128\SECURED_DRIVE\Secret Project Data\pricing decision\secret_p...	2015-03-23 20:26:53	informant
Jump list	\\10.11.11.128\SECURED_DRIVE\Secret Project Data\pricing decision	2015-03-23 20:26:53	informant
IE / WinInet history (file opened)	\\10.11.11.128\secured_drive\Secret Project Data\pricing decision\secret_proj...	2015-03-23 20:26:54	informant
Shortcut (LNK)	\\10.11.11.128\SECURED_DRIVE\Secret Project Data\pricing decision	2015-03-23 20:26:54	informant
Shortcut (LNK)	\\10.11.11.128\SECURED_DRIVE\Secret Project Data\pricing decision\secret_p...	2015-03-23 20:26:56	informant
Office MRU	\\10.11.11.128\secured_drive\Secret Project Data\pricing decision\secret_proj...	2015-03-23 20:26:56	informant
Office MRU	\\10.11.11.128\secured_drive\Secret Project Data\pricing decision\	2015-03-23 20:26:56	informant
Shellbag (folder browsed)	V:	2015-03-23 20:27:24	informant
Shellbag (folder browsed)	V:\Secret Project Data	2015-03-23 20:27:29	informant
Shellbag (folder browsed)	V:\Secret Project Data\final	2015-03-23 20:27:29	informant
IE / WinInet history (file opened)	V:\Secret Project Data\final\secret_project_final_meeting.pptx	2015-03-23 20:27:33	informant
Shortcut (LNK)	V:\Secret Project Data\final\secret_project_final_meeting.pptx	2015-03-23 20:27:33	informant
Shortcut (LNK)	V:\Secret Project Data\final	2015-03-23 20:27:33	informant
Jump list	V:\Secret Project Data\final\secret_project_final_meeting.pptx	2015-03-23 20:27:33	informant
Jump list	V:\Secret Project Data\final	2015-03-23 20:27:33	informant
Shortcut (LNK)	V:\Secret Project Data\final\secret_project_final_meeting.pptx	2015-03-23 20:27:37	informant
Office MRU	V:\Secret Project Data\final\secret_project_final_meeting.pptx	2015-03-23 20:27:37	informant
Office MRU	V:\Secret Project Data\final\	2015-03-23 20:27:37	informant

22 rows | 1 highlighted

1 File / folder on a network share

Figure 27: Network share activity - Suspect workstation (INFORMANT-PC)

Supporting artifacts

Artifact	Time (UTC)	Summary	Source
Mapped Network Drives	2015-03-23 20:26:04	Mapped network drive \\10.11.11.128\secured_drive	NTUSER\...\Map Network Drive MRU
User Mounted Volumes (MountPoints2)	2015-03-23 20:26:04	informant mounted ##10.11.11.128#secured_drive (network share)	NTUSER\Software\Microsoft\Windows\CurrentVersion\Explorer\MountPoints2
Files Opened (IE / WinInet file:/// history)	2015-03-23 20:26:53	Opened \\10.11.11.128\secured_drive\Secret Project Data\pricing decision\secret_project_pricing_decision.xlsx	C:\Users\informant\AppData\Local\Microsoft\Windows\WebCache\WebCacheV01.dat
Files Opened (IE / WinInet file:/// history)	2015-03-23 20:26:54	Opened \\10.11.11.128\secured_drive\Secret Project Data\pricing decision\secret_project_pricing_decision.xlsx	C:\Users\informant\AppData\Local\Microsoft\Windows\WebCache\WebCacheV01.dat
Files Opened (IE / WinInet file:/// history)	2015-03-23 20:27:33	Opened V:\Secret Project Data\final\secret_project_final_meeting.pptx	C:\Users\informant\AppData\Local\Microsoft\Windows\WebCache\WebCacheV01.dat
Shortcut (LNK) Files	2015-03-23 20:26:53	LNK -> \\10.11.11.128\SECURED_DRIVE\Secret Project Data\pricing decision\secret_project_pricing_decision.xlsx (NETWORK)	C:\Users\informant\AppData\Roaming\Microsoft\Windows\Recent\secret_project_pricing_decision.xlsx.lnk
Shortcut (LNK) Files	2015-03-23 20:26:54	LNK -> \\10.11.11.128\SECURED_DRIVE\Secret Project Data\pricing decision (NETWORK)	C:\Users\informant\AppData\Roaming\Microsoft\Windows\Recent\pricing decision.lnk
Shortcut (LNK) Files	2015-03-23 20:26:56	LNK -> \\10.11.11.128\SECURED_DRIVE\Secret Project Data\pricing decision\secret_project_pricing_decision.xlsx (NETWORK)	C:\Users\informant\AppData\Roaming\Microsoft\Office\Recent\secret_project_pricing_decision.xlsx.LNK

F-16 Consolidated timeline of key events

System: Case-wide	Time: -	Supporting artifacts: 0 Timeline
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870 events were placed on the case timeline (148 flagged as relevant by the analyzers) for the period 2009-07-14 00:34:08 to 2015-03-25 17:31:00 UTC. The full timeline is available in the Excel workbook and in the application.

Timeline of key events per system

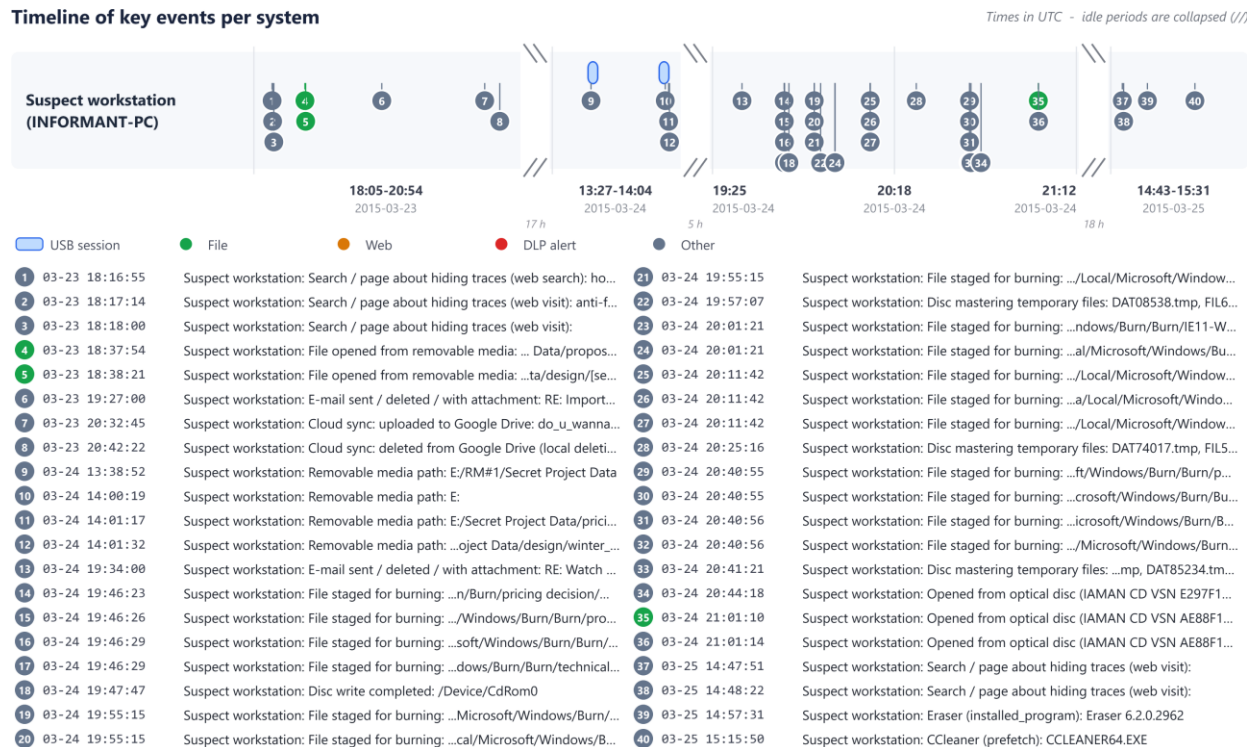


Figure 28: Timeline of key events per system

Consolidated timeline of key events

Time (UTC)	System	Source	Event	Description
2015-03-23 18:16:55	Suspect workstation (INF...	Anti-forensics	Search / page about hiding ...	how to delete data
2015-03-23 18:16:55	Suspect workstation (INF...	Anti-forensics	Search / page about hiding ...	how to delete data - Google Search
2015-03-23 18:17:14	Suspect workstation (INF...	Anti-forensics	Search / page about hiding ...	anti-forensics
2015-03-23 18:17:14	Suspect workstation (INF...	Anti-forensics	Search / page about hiding ...	anti-forensics - Google Search
2015-03-23 18:17:19	Suspect workstation (INF...	Anti-forensics	Search / page about hiding ...	
2015-03-23 18:17:19	Suspect workstation (INF...	Anti-forensics	Search / page about hiding ...	Anti-forensic techniques - ForensicsWiki
2015-03-23 18:17:57	Suspect workstation (INF...	Anti-forensics	Search / page about hiding ...	
2015-03-23 18:18:00	Suspect workstation (INF...	Anti-forensics	Search / page about hiding ...	
2015-03-23 18:31:09	Suspect workstation (INF...	USB	USB first seen	SanDisk Cruzer Fit S/N 4C530012450531101593 first connected
2015-03-23 18:37:20	Suspect workstation (INF...	LNK	File opened from removabl...	E:\RM#1\Secret Project Data\proposal\[secret_project]_proposal.d...
2015-03-23 18:37:54	Suspect workstation (INF...	LNK	File opened from removabl...	E:\RM#1\Secret Project Data\proposal\[secret_project]_proposal.d...
2015-03-23 18:37:54	Suspect workstation (INF...	Jump list	File opened from removabl...	E:\RM#1\Secret Project Data\proposal (VSN 5C754D3E, label Author...
2015-03-23 18:38:21	Suspect workstation (INF...	Jump list	File opened from removabl...	E:\RM#1\Secret Project Data\design\[secret_project]_design_conce...
2015-03-23 18:38:21	Suspect workstation (INF...	Jump list	File opened from removabl...	E:\ (VSN 5C754D3E, label Authorized USB)
2015-03-23 18:38:21	Suspect workstation (INF...	LNK	File opened from removabl...	E:\RM#1\Secret Project Data\design\[secret_project]_design_conce...
2015-03-23 18:38:23	Suspect workstation (INF...	Jump list	File opened from removabl...	E:\RM#1\Secret Project Data\design (VSN 5C754D3E, label Authoriz...
2015-03-23 18:38:23	Suspect workstation (INF...	LNK	File opened from removabl...	E:\RM#1\Secret Project Data\design\[secret_project]_design_conce...
2015-03-23 18:44:00	Suspect workstation (INF...	E-mail	E-mail sent / deleted / with ...	RE: Hello, iaman - iaman (Exchange mailbox) -> spy
2015-03-23 19:27:00	Suspect workstation (INF...	E-mail	E-mail sent / deleted / with ...	RE: Important request - iaman (Exchange mailbox) -> spy
2015-03-23 20:06:08	Suspect workstation (INF...	Google Drive (leg...	Cloud sync: deleted from G...	How to get started with Drive
2015-03-23 20:32:45	Suspect workstation (INF...	Google Drive (leg...	Cloud sync: uploaded to Go...	happy_holiday.jpg
2015-03-23 20:32:45	Suspect workstation (INF...	Google Drive (leg...	Cloud sync: uploaded to Go...	do_u_wanna_build_a_snow_man.mp3

... 38 more rows - see the Excel workbook / Parsed CSV

Figure 29: Consolidated timeline of key events

6. Timeline of key events

The table lists the 148 events flagged by the analyzers in chronological order (UTC, with local time in America/New_York). The complete timeline is in the accompanying Excel workbook.

UTC	Local	System	Source	Event	Description
2015-03-23 18:16:55	2015-03-23 14:16:55	Suspect workstation	Anti-forensics	Search / page about hiding traces (web search)	how to delete data
2015-03-23 18:16:55	2015-03-23 14:16:55	Suspect workstation	Anti-forensics	Search / page about hiding traces (web visit)	how to delete data - Google Search
2015-03-23 18:17:14	2015-03-23 14:17:14	Suspect workstation	Anti-forensics	Search / page about hiding traces (web search)	anti-forensics
2015-03-23 18:17:14	2015-03-23 14:17:14	Suspect workstation	Anti-forensics	Search / page about hiding traces (web visit)	anti-forensics - Google Search
2015-03-23 18:17:19	2015-03-23 14:17:19	Suspect workstation	Anti-forensics	Search / page about hiding traces (web visit)	
2015-03-23 18:17:19	2015-03-23 14:17:19	Suspect workstation	Anti-forensics	Search / page about hiding traces (web visit)	Anti-forensic techniques - ForensicsWiki
2015-03-23 18:17:57	2015-03-23 14:17:57	Suspect workstation	Anti-forensics	Search / page about hiding traces (web visit)	
2015-03-23 18:18:00	2015-03-23 14:18:00	Suspect workstation	Anti-forensics	Search / page about hiding traces (web visit)	
2015-03-23 18:31:09	2015-03-23 14:31:09	Suspect workstation	USB	USB first seen	SanDisk Cruzer Fit S/N 4C530012450531101593 first connected
2015-03-23 18:37:20	2015-03-23 14:37:20	Suspect workstation	LNK	File opened from removable media	E:\RM#1\Secret Project Data\proposal\[secret_project]_proposal.docx (VSN 5C754D3E, label Authorized USB)
2015-03-23 18:37:54	2015-03-23 14:37:54	Suspect workstation	LNK	File opened from removable media	E:\RM#1\Secret Project Data\proposal\[secret_project]_proposal.docx (VSN 5C754D3E, label Authorized USB)
2015-03-23 18:37:54	2015-03-23 14:37:54	Suspect workstation	Jump list	File opened from removable media	E:\RM#1\Secret Project Data\design (VSN 5C754D3E, label Authorized USB)
2015-03-23 18:38:21	2015-03-23 14:38:21	Suspect workstation	Jump list	File opened from removable media	E:\RM#1\Secret Project Data\design\[secret_project]_design_concept.ppt (VSN 5C754D3E, label Authorized USB)
2015-03-23 18:38:21	2015-03-23 14:38:21	Suspect workstation	Jump list	File opened from removable media	E:\(VSN 5C754D3E, label Authorized USB)
2015-03-23 18:38:21	2015-03-23 14:38:21	Suspect workstation	LNK	File opened from removable media	E:\RM#1\Secret Project Data\design\[secret_project]_design_concept.ppt (VSN 5C754D3E, label Authorized USB)
2015-03-23 18:38:23	2015-03-23 14:38:23	Suspect workstation	Jump list	File opened from removable media	E:\RM#1\Secret Project Data\design (VSN 5C754D3E, label Authorized USB)
2015-03-23 18:38:23	2015-03-23 14:38:23	Suspect workstation	LNK	File opened from removable media	E:\RM#1\Secret Project Data\design\[secret_project]_design_concept.ppt (VSN 5C754D3E, label Authorized USB)
2015-03-23 18:44:00	2015-03-23 14:44:00	Suspect workstation	E-mail	E-mail sent / deleted / with attachment	RE: Hello, Iaman - iaman (Exchange mailbox) -> spy
2015-03-23 19:27:00	2015-03-23 15:27:00	Suspect workstation	E-mail	E-mail sent / deleted / with attachment	RE: Important request - iaman (Exchange mailbox) -> spy
2015-03-23 20:06:08	2015-03-23 16:06:08	Suspect workstation	Google Drive (legacy)	Cloud sync: deleted from Google Drive (local deletion synchronized)	How to get started with Drive
2015-03-23 20:32:45	2015-03-23 16:32:45	Suspect workstation	Google Drive (legacy)	Cloud sync: uploaded to Google Drive	happy_holiday.jpg
2015-03-23 20:32:45	2015-03-23 16:32:45	Suspect workstation	Google Drive (legacy)	Cloud sync: uploaded to Google Drive	do_u_wanna_build_a_snow_man.mp3
2015-03-23 20:38:47	2015-03-23 16:38:47	Suspect workstation	E-mail	E-mail sent / deleted / with attachment	It's me - -> spy.conspirator@nist.gov
2015-03-23 20:41:22	2015-03-23 16:41:22	Suspect workstation	E-mail	E-mail sent / deleted / with attachment	RE: It's me - spy <spy.conspirator@nist.gov> -> iaman <iaman.informant@nist.gov>
2015-03-23 20:42:20	2015-03-23 16:42:20	Suspect workstation	Google Drive (legacy)	Cloud sync: deleted from Google Drive (local deletion synchronized)	happy_holiday.jpg
2015-03-23 20:42:22	2015-03-23 16:42:22	Suspect workstation	Google Drive (legacy)	Cloud sync: deleted from Google Drive (local deletion synchronized)	do_u_wanna_build_a_snow_man.mp3

UTC	Local	System	Source	Event	Description
2015-03-24 13:30:10	2015-03-24 09:30:10	Suspect workstation	E-mail	E-mail sent / deleted / with attachment	RE: Last request - -> spy.conspirator@nist.gov
2015-03-24 13:35:00	2015-03-24 09:35:00	Suspect workstation	E-mail	E-mail sent / deleted / with attachment	RE: Last request - iaman (Exchange mailbox) -> spy
2015-03-24 13:38:00	2015-03-24 09:38:00	Suspect workstation	USB	USB connected	SanDisk Cruzer Fit S/N 4C530012450531101593 connected
2015-03-24 13:38:31	2015-03-24 09:38:31	Suspect workstation	Shellbag (folder browsed)	Removable media path	E:\RM#1
2015-03-24 13:38:52	2015-03-24 09:38:52	Suspect workstation	Shellbag (folder browsed)	Removable media path	E:\RM#1\Secret Project Data
2015-03-24 13:38:52	2015-03-24 09:38:52	Suspect workstation	Shellbag (folder browsed)	Removable media path	E:\RM#1\Secret Project Data\design
2015-03-24 13:58:31	2015-03-24 09:58:31	Suspect workstation	USB	USB first seen	SanDisk Cruzer Fit S/N 4C530012550531106501 first connected
2015-03-24 13:58:33	2015-03-24 09:58:33	Suspect workstation	USB	USB connected	SanDisk Cruzer Fit S/N 4C530012550531106501 connected (E: IAMAN \$ _@)
2015-03-24 14:00:19	2015-03-24 10:00:19	Suspect workstation	Shellbag (folder browsed)	Removable media path	E:
2015-03-24 14:01:11	2015-03-24 10:01:11	Suspect workstation	Shellbag (folder browsed)	Removable media path	E:\Secret Project Data\technical review
2015-03-24 14:01:14	2015-03-24 10:01:14	Suspect workstation	Shellbag (folder browsed)	Removable media path	E:\Secret Project Data\proposal
2015-03-24 14:01:15	2015-03-24 10:01:15	Suspect workstation	Shellbag (folder browsed)	Removable media path	E:\Secret Project Data\progress
2015-03-24 14:01:17	2015-03-24 10:01:17	Suspect workstation	Shellbag (folder browsed)	Removable media path	E:\Secret Project Data\pricing decision
2015-03-24 14:01:23	2015-03-24 10:01:23	Suspect workstation	Jump list	File opened from removable media	E:\Secret Project Data\design\winter_whether_advisory.zip (VSN 9E6A5B82, label IAMAN \$ _@)
2015-03-24 14:01:29	2015-03-24 10:01:29	Suspect workstation	Shellbag (folder browsed)	Removable media path	E:\Secret Project Data\design
2015-03-24 14:01:29	2015-03-24 10:01:29	Suspect workstation	Jump list	File opened from removable media	E:\SECRET~1\design\WINTER~1.ZIP (VSN , label)
2015-03-24 14:01:32	2015-03-24 10:01:32	Suspect workstation	Shellbag (folder browsed)	Removable media path	E:\Secret Project Data\design\winter_whether_advisory.zip
2015-03-24 14:01:32	2015-03-24 10:01:32	Suspect workstation	Shellbag (folder browsed)	Removable media path	E:\Secret Project Data\design\winter_whether_advisory.zip<UNK NOWN size=0x006a type=None>
2015-03-24 19:32:42	2015-03-24 15:32:42	Suspect workstation	E-mail	E-mail sent / deleted / with attachment	Watch out! - 16DEB821A6D2423FB723680AE0D2B36A-SPY -> iaman.informant@nist.gov
2015-03-24 19:34:00	2015-03-24 15:34:00	Suspect workstation	E-mail	E-mail sent / deleted / with attachment	RE: Watch out! - iaman (Exchange mailbox) -> spy
2015-03-24 19:46:16	2015-03-24 15:46:16	Suspect workstation	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\design\winter_storm.amr
2015-03-24 19:46:19	2015-03-24 15:46:19	Suspect workstation	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\design\winter_whether_advisory.zip
2015-03-24 19:46:23	2015-03-24 15:46:23	Suspect workstation	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\pricing decision\my_favorite_cars.db
2015-03-24 19:46:23	2015-03-24 15:46:23	Suspect workstation	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\pricing decision\my_favorite_movies.7z
2015-03-24 19:46:23	2015-03-24 15:46:23	Suspect workstation	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\pricing decision\new_years_day.jpg
2015-03-24 19:46:25	2015-03-24 15:46:25	Suspect workstation	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\pricing decision\super_bowl.avi
2015-03-24 19:46:26	2015-03-24 15:46:26	Suspect workstation	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\progress\my_friends.svg

UTC	Local	System	Source	Event	Description
2015-03-24 19:46:26	2015-03-24 15:46:26	Suspect workstation	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\progress\my_smartphone.png
2015-03-24 19:46:27	2015-03-24 15:46:27	Suspect workstation	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\progress\new_year_calendar.one
2015-03-24 19:46:27	2015-03-24 15:46:27	Suspect workstation	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\proposal\gift_from_you.gif
2015-03-24 19:46:29	2015-03-24 15:46:29	Suspect workstation	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\proposal\landscape.png
2015-03-24 19:46:29	2015-03-24 15:46:29	Suspect workstation	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\technical review\diary_#1d.txt
2015-03-24 19:46:29	2015-03-24 15:46:29	Suspect workstation	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\technical review\diary_#1p.txt
2015-03-24 19:46:29	2015-03-24 15:46:29	Suspect workstation	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\technical review\diary_#2d.txt
2015-03-24 19:46:29	2015-03-24 15:46:29	Suspect workstation	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\technical review\diary_#2p.txt
2015-03-24 19:46:29	2015-03-24 15:46:29	Suspect workstation	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\technical review\diary_#3d.txt
2015-03-24 19:46:29	2015-03-24 15:46:29	Suspect workstation	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\technical review\diary_#3p.txt
2015-03-24 19:47:47	2015-03-24 15:47:47	Suspect workstation	\$UsnJrnl	Disc mastering temporary files	DAT34216.tmp, DAT67383.tmp, FIL39751.tmp, POST39751.tmp
2015-03-24 19:47:47	2015-03-24 15:47:47	Suspect workstation	System event log (cdrom 133)	Disc write completed	\Device\CdRom0
2015-03-24 19:49:26	2015-03-24 15:49:26	Suspect workstation	\$UsnJrnl	Disc mastering temporary files	DAT34216.tmp, FIL39751.tmp, POST39751.tmp
2015-03-24 19:54:35	2015-03-24 15:54:35	Suspect workstation	Shellbag	Explorer disc-burning view opened	D:\<CDBURN>
2015-03-24 19:55:15	2015-03-24 15:55:15	Suspect workstation	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\Chrysanthemum.jpg
2015-03-24 19:55:15	2015-03-24 15:55:15	Suspect workstation	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\Desert.jpg
2015-03-24 19:55:15	2015-03-24 15:55:15	Suspect workstation	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\Hydrangeas.jpg
2015-03-24 19:55:15	2015-03-24 15:55:15	Suspect workstation	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\IE11-Windows6.1-x64-en-us.exe
2015-03-24 19:55:15	2015-03-24 15:55:15	Suspect workstation	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\Jellyfish.jpg
2015-03-24 19:55:15	2015-03-24 15:55:15	Suspect workstation	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\Koala.jpg
2015-03-24 19:55:15	2015-03-24 15:55:15	Suspect workstation	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\Lighthouse.jpg
2015-03-24 19:55:15	2015-03-24 15:55:15	Suspect workstation	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\Penguins.jpg
2015-03-24 19:55:15	2015-03-24 15:55:15	Suspect workstation	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\Tulips.jpg
2015-03-24 19:56:01	2015-03-24 15:56:01	Suspect workstation	\$UsnJrnl	Disc mastering temporary files	DAT08538.tmp, DAT32224.tmp, FIL66692.tmp, POST66692.tmp
2015-03-24 19:56:11	2015-03-24 15:56:11	Suspect workstation	System event log (cdrom 133)	Disc write completed	\Device\CdRom0
2015-03-24 19:57:07	2015-03-24 15:57:07	Suspect workstation	\$UsnJrnl	Disc mastering temporary files	DAT08538.tmp, FIL66692.tmp, POST66692.tmp

UTC	Local	System	Source	Event	Description
2015-03-24 20:01:21	2015-03-24 16:01:21	Suspect workstation	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\Chrysanthemum.jpg
2015-03-24 20:01:21	2015-03-24 16:01:21	Suspect workstation	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\Desert.jpg
2015-03-24 20:01:21	2015-03-24 16:01:21	Suspect workstation	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\Hydrangeas.jpg
2015-03-24 20:01:21	2015-03-24 16:01:21	Suspect workstation	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\IE11-Windows6.1-x64-en-us.exe
2015-03-24 20:01:21	2015-03-24 16:01:21	Suspect workstation	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\Jellyfish.jpg
2015-03-24 20:01:21	2015-03-24 16:01:21	Suspect workstation	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\Koala.jpg
2015-03-24 20:01:21	2015-03-24 16:01:21	Suspect workstation	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\Lighthouse.jpg
2015-03-24 20:01:21	2015-03-24 16:01:21	Suspect workstation	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\Penguins.jpg
2015-03-24 20:01:21	2015-03-24 16:01:21	Suspect workstation	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\Tulips.jpg
2015-03-24 20:11:42	2015-03-24 16:11:42	Suspect workstation	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\Chrysanthemum.jpg
2015-03-24 20:11:42	2015-03-24 16:11:42	Suspect workstation	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\Desert.jpg
2015-03-24 20:11:42	2015-03-24 16:11:42	Suspect workstation	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\Hydrangeas.jpg
2015-03-24 20:11:42	2015-03-24 16:11:42	Suspect workstation	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\IE11-Windows6.1-x64-en-us.exe
2015-03-24 20:11:42	2015-03-24 16:11:42	Suspect workstation	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\Jellyfish.jpg
2015-03-24 20:11:42	2015-03-24 16:11:42	Suspect workstation	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\Koala.jpg
2015-03-24 20:11:42	2015-03-24 16:11:42	Suspect workstation	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\Lighthouse.jpg
2015-03-24 20:11:42	2015-03-24 16:11:42	Suspect workstation	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\Penguins.jpg
2015-03-24 20:11:42	2015-03-24 16:11:42	Suspect workstation	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\Tulips.jpg
2015-03-24 20:24:14	2015-03-24 16:24:14	Suspect workstation	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\Tulips.jpg
2015-03-24 20:24:19	2015-03-24 16:24:19	Suspect workstation	\$UsnJrnl	Disc mastering temporary files	DAT67829.tmp, DAT74017.tmp, FIL51898.tmp, POST51898.tmp
2015-03-24 20:24:46	2015-03-24 16:24:46	Suspect workstation	System event log (cdrom 133)	Disc write completed	\Device\CdRom0
2015-03-24 20:25:16	2015-03-24 16:25:16	Suspect workstation	\$UsnJrnl	Disc mastering temporary files	DAT74017.tmp, FIL51898.tmp, POST51898.tmp
2015-03-24 20:40:55	2015-03-24 16:40:55	Suspect workstation	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\de\winter_storm.amr
2015-03-24 20:40:55	2015-03-24 16:40:55	Suspect workstation	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\de\winter_weather_advisory.zip
2015-03-24 20:40:55	2015-03-24 16:40:55	Suspect workstation	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\pd\my_favorite_cars.db

UTC	Local	System	Source	Event	Description
2015-03-24 20:40:55	2015-03-24 16:40:55	Suspect workstation	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\pd\my_favorite_movies.7z
2015-03-24 20:40:55	2015-03-24 16:40:55	Suspect workstation	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\pd\new_years_day.jpg
2015-03-24 20:40:55	2015-03-24 16:40:55	Suspect workstation	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\pd\super_bowl.avi
2015-03-24 20:40:55	2015-03-24 16:40:55	Suspect workstation	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\prog\my_friends.svg
2015-03-24 20:40:55	2015-03-24 16:40:55	Suspect workstation	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\prog\my_smartphone.png
2015-03-24 20:40:55	2015-03-24 16:40:55	Suspect workstation	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\prog\new_year_calendar.one
2015-03-24 20:40:55	2015-03-24 16:40:55	Suspect workstation	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\prop\gift_from_you.gif
2015-03-24 20:40:56	2015-03-24 16:40:56	Suspect workstation	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\prop\landscape.png
2015-03-24 20:40:56	2015-03-24 16:40:56	Suspect workstation	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\tr\diary_#1d.txt
2015-03-24 20:40:56	2015-03-24 16:40:56	Suspect workstation	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\tr\diary_#1p.txt
2015-03-24 20:40:56	2015-03-24 16:40:56	Suspect workstation	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\tr\diary_#2d.txt
2015-03-24 20:40:56	2015-03-24 16:40:56	Suspect workstation	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\tr\diary_#2p.txt
2015-03-24 20:40:56	2015-03-24 16:40:56	Suspect workstation	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\tr\diary_#3d.txt
2015-03-24 20:40:56	2015-03-24 16:40:56	Suspect workstation	Burn staging folder (\$UsnJrnl)	File staged for burning	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\tr\diary_#3p.txt
2015-03-24 20:41:21	2015-03-24 16:41:21	Suspect workstation	\$UsnJrnl	Disc mastering temporary files	DAT11399.tmp, DAT85234.tmp, FIL61821.tmp, POST61821.tmp
2015-03-24 20:41:21	2015-03-24 16:41:21	Suspect workstation	System event log (cdrom 133)	Disc write completed	\Device\CdRom0
2015-03-24 20:43:10	2015-03-24 16:43:10	Suspect workstation	\$UsnJrnl	Disc mastering temporary files	DAT11399.tmp, FIL61821.tmp, POST61821.tmp
2015-03-24 20:44:13	2015-03-24 16:44:13	Suspect workstation	Jump list	Opened from optical disc (IAMAN CD VSN E297F1A5)	D:\de\winter_whether_advisory.zip
2015-03-24 20:44:18	2015-03-24 16:44:18	Suspect workstation	Shortcut (LNK)	Opened from optical disc (IAMAN CD VSN E297F1A5)	D:\de\winter_whether_advisory.zip
2015-03-24 20:47:30	2015-03-24 16:47:30	Suspect workstation	Shortcut (LNK)	Opened from optical disc (IAMAN CD VSN A58AD72A)	D:\
2015-03-24 20:53:16	2015-03-24 16:53:16	Suspect workstation	Registry (CD Burning)	Burn method last selected	DefaultToMastered = 0 (Live File System - like a USB flash drive)
2015-03-24 20:54:07	2015-03-24 16:54:07	Suspect workstation	Shellbag (folder browsed)	Removable media path	E:\Secret Project Data
2015-03-24 21:01:10	2015-03-24 17:01:10	Suspect workstation	Jump list	Opened from optical disc (IAMAN CD VSN AE88F176)	D:\
2015-03-24 21:01:10	2015-03-24 17:01:10	Suspect workstation	Shortcut (LNK)	Opened from optical disc (IAMAN CD VSN AE88F176)	D:\Penguins.jpg
2015-03-24 21:01:12	2015-03-24 17:01:12	Suspect workstation	Shortcut (LNK)	Opened from optical disc (IAMAN CD VSN AE88F176)	D:\Koala.jpg
2015-03-24 21:01:14	2015-03-24 17:01:14	Suspect workstation	Shortcut (LNK)	Opened from optical disc (IAMAN CD VSN AE88F176)	D:\Tulips.jpg

UTC	Local	System	Source	Event	Description
2015-03-24 21:01:14	2015-03-24 17:01:14	Suspect workstation	Shortcut (LNK)	Opened from optical disc (IAMAN CD VSN AE88F176)	D:\
2015-03-24 21:05:00	2015-03-24 17:05:00	Suspect workstation	E-mail	E-mail sent / deleted / with attachment	Done - iaman (Exchange mailbox) -> spy
2015-03-25 14:46:44	2015-03-25 10:46:44	Suspect workstation	Anti-forensics	Search / page about hiding traces (web visit)	
2015-03-25 14:47:51	2015-03-25 10:47:51	Suspect workstation	Anti-forensics	Search / page about hiding traces (web visit)	
2015-03-25 14:47:55	2015-03-25 10:47:55	Suspect workstation	Anti-forensics	Search / page about hiding traces (web visit)	
2015-03-25 14:48:12	2015-03-25 10:48:12	Suspect workstation	Anti-forensics	Search / page about hiding traces (web visit)	
2015-03-25 14:48:21	2015-03-25 10:48:21	Suspect workstation	Anti-forensics	Search / page about hiding traces (web visit)	
2015-03-25 14:48:22	2015-03-25 10:48:22	Suspect workstation	Anti-forensics	Search / page about hiding traces (web visit)	
2015-03-25 14:48:22	2015-03-25 10:48:22	Suspect workstation	Anti-forensics	Search / page about hiding traces (web visit)	
2015-03-25 14:48:39	2015-03-25 10:48:39	Suspect workstation	Anti-forensics	Search / page about hiding traces (web visit)	
2015-03-25 14:57:31	2015-03-25 10:57:31	Suspect workstation	Anti-forensics	Eraser (installed_program)	Eraser 6.2.0.2962
2015-03-25 15:12:28	2015-03-25 11:12:28	Suspect workstation	Anti-forensics	Eraser (userassist)	C:\Program Files\Eraser\Eraser.exe
2015-03-25 15:13:30	2015-03-25 11:13:30	Suspect workstation	Anti-forensics	Eraser (prefetch)	ERASER.EXE
2015-03-25 15:15:50	2015-03-25 11:15:50	Suspect workstation	Anti-forensics	CCleaner (userassist)	C:\Program Files\CCleaner\CCleaner64.exe
2015-03-25 15:15:50	2015-03-25 11:15:50	Suspect workstation	Anti-forensics	CCleaner (prefetch)	CLEANER64.EXE
2015-03-25 15:21:30	2015-03-25 11:21:30	Suspect workstation	userassist	Google Drive / Backup and Sync (legacy client) executed	C:\Program Files (x86)\Google\Drive\googledrivesync.exe
2015-03-25 15:21:31	2015-03-25 11:21:31	Suspect workstation	prefetch	Google Drive / Backup and Sync (legacy client) executed	GOOGLEDRIVESYNC.EXE
2015-03-25 15:28:33	2015-03-25 11:28:33	Suspect workstation	Registry	Browser file dialog	iexplore.exe used Download\ ()

7. Keyword and target-file search results

7.1 Suspect workstation (INFORMANT-PC)

Search term	Origin	Hits	Hits by area
secret_project	keyword	525	\$UsnJrnl (change journal) [C:]: 210; Allocated file [C:]: 119; Volume Shadow Copy store [C:]: 89; Unallocated space [C:]: 26; Unallocated (deleted file remnant) [C:]: 22; pagefile.sys [C:]: 21; MFT record (resident data)
spy.conspirator	keyword	5	Allocated file [C:]: 4; File slack [C:]: 1
confidential	keyword	2234	Allocated file [C:]: 2112; Unallocated space [C:]: 59; File slack [C:]: 36; pagefile.sys [C:]: 15; Unallocated (deleted file remnant) [C:]: 11; Volume Shadow Copy store [C:]: 1

8. Artifact coverage - every location examined

This matrix documents every artifact source the profile examined on each evidence item and the result, so that a 'no evidence found' conclusion can be read together with the locations that were actually checked.

8.1 Suspect workstation (INFORMANT-PC)

Artifact	Location checked	Result	Records	Notes
USN change journal	Vol1 \$Extend\\$UsnJrnl:\$J	Not present on system		journal not present / disabled
MFT (incl. deleted records)	Vol1 \$MFT	Found	56	0 deleted file records

Artifact	Location checked	Result	Records	Notes
Zone.Identifier streams	Vol1 alternate data streams	Checked - nothing found		
Timestamp anomalies	Vol1 \$\$I vs \$FN	Checked - nothing found		
USN change journal	C: \$Extend\UsnJrnl:\$J	Found	317137	
MFT (incl. deleted records)	C: \$MFT	Found	59943	911 deleted file records
Zone.Identifier streams	C: alternate data streams	Found	6	
Timestamp anomalies	C: \$\$I vs \$FN	Found	7844	
Logon / Logoff Events	User Profile Service/Operational, Security	Found	309	
Remote Desktop / Remote Session Events	Terminalsrvcs-LocalSessionManager/Operational, Security	Found	35	
Process Execution Events	Security	Checked - nothing found		
Service Events	Security, System	Found	89	
Scheduled Task Events	Security	Checked - nothing found		
PowerShell Events	Powershell/Operational, Powershellcore/Operational, Windows Powershell	Not present on system		none of the source logs exist on this system
Microsoft Defender Events	Application	Checked - nothing found		
Account Management Events	Security	Found	62	
Log Clearing / Audit Policy Events	Security, System	Checked - nothing found		
System Power / Time Events	Security, System	Found	79	
Network Events	Bits-Client/Operational, Networkprofile/Operational, Security, System	Found	145	
Application Events	Application, Oalerts	Found	127	
Print Jobs	Printservice/Operational	Not present on system		none of the source logs exist on this system
Optical Disc Write Events	System	Found	4	
Network Share Events	Security	Checked - nothing found		
File Object Access (Audit)	Security	Checked - nothing found		
Sysmon Events	Sysmon/Operational	Not present on system		none of the source logs exist on this system
Persistence / Configuration Events	Windows Firewall With Advanced Security/Firewall, Security	Found	56	
USB / device events	Partition, Kernel-PnP, DriverFrameworks, StorSvc, Ntfs, VHDMP, Security 6416	Found	26	
Prefetch	C:\Windows\Prefetch*.pf	Found	95	
Amcache	C:\Windows\AppCompat\Programs\Amcache.hve	Not present on system		
Program Compatibility Assistant	C:\Windows\appcompat\pca	Not present on system		
Shimcache	SYSTEM\ControlSet00x\Control\Session Manager\AppCompatCache	Found	305	
BAM / DAM	SYSTEM\...\Services\bam\State\UserSettings	Checked - nothing found		
SRUM	C:\Windows\System32\sr\SRUDB.dat	Not present on system		
MUICache	UsrClass.dat\...\Shell\MuiCache	Found	5	
RecentDocs	NTUSER\...\Explorer\RecentDocs	Found	34	
Open/Save dialog MRU	NTUSER\...\ComDlg32\OpenSavePidMRU	Found	10	
Last visited folder MRU	NTUSER\...\ComDlg32\LastVisitedPidMRU	Found	3	
Run box history (RunMRU)	NTUSER\...\Explorer\RunMRU	Found	2	
Explorer typed paths	NTUSER\...\Explorer\TypedPaths	Checked - nothing found		
Explorer search terms	NTUSER\...\Explorer\WordWheelQuery	Found	1	
UserAssist	NTUSER\...\Explorer\UserAssist	Found	105	
Office file MRU	NTUSER\Software\Microsoft\Office\<ver>\<app>\File MRU	Found	10	
Office trusted documents	NTUSER\...\Trusted Documents\TrustRecords	Checked - nothing found		
Typed URLs	NTUSER\...\Internet Explorer\TypedURLs	Found	5	
RDP client history	NTUSER\...\Terminal Server Client	Checked - nothing found		

Artifact	Location checked	Result	Records	Notes
Mapped network drives	NTUSER\...\Map Network Drive MRU, NTUSER\Network	Found	1	
Shellbags	UsrClass.dat / NTUSER.DAT BagMRU	Found	113	
Browser extensions	User Data\<profile>\Extensions*\manifest.json	Found	22	
Chromium browser history	User Data\<profile>\History	Found	2	Chrome
Firefox history	Profiles*\places.sqlite	Not present on system		
Web visit	browser profiles	Found	145	
Web download	browser profiles	Found	2	
Web search	browser profiles	Found	29	
Web login	browser profiles	Checked - nothing found		
Web autofill	browser profiles	Checked - nothing found		
Web cookie host	browser profiles	Found	83	
Shortcut (LNK) files	all *.lnk (incl. deleted MFT records)	Found	226	10 recovered from deleted records
Jump lists	Recent\AutomaticDestinations + CustomDestinations	Found	104	
Recycle Bin (\$I files)	<drive>\\$Recycle.Bin\<SID>	Found	14	
Windows Timeline	ConnectedDevicesPlatform*\ActivitiesCache.db	Not present on system		
IE / Edge legacy WebCache	Users*\AppData\Local\Microsoft\Windows\WebCache\WebCacheV01.dat	Found	274	3 database(s)
IE 5-9 history (index.dat)	Users*\AppData\Local\Microsoft\Windows\History\History.IE5**\index.dat	Checked - nothing found		2 file(s)
Wi-Fi profiles	ProgramData\Microsoft\Wlansvc\Profiles	Checked - nothing found		
Hosts / proxy / VPN settings	hosts, Internet Settings, rasphone.pbk	Found	1	
PowerShell console history	...\PSReadLine\ConsoleHost_history.txt	Not present on system		
PowerShell transcripts	PowerShell_transcript.*.txt	Checked - nothing found		
Scripts / containers in user-writable folders	Temp, Downloads, AppData, ProgramData, Public, Desktop	Found	474	
USB device enumeration	SYSTEM\Enum\USBSTOR, SCSI, USB, SWD\WPDBUSENUM	Found	8	
DeviceClasses (interface keys)	SYSTEM\CurrentControlSet\Control\DeviceClasses	Found	16	
MountedDevices	SYSTEM\MountedDevices	Found	10	
EMDMgmt (ReadyBoost volume history)	SOFTWARE\...\EMDMgmt	Checked - nothing found		
Windows Portable Devices	SOFTWARE\Microsoft\Windows Portable Devices\Devices	Checked - nothing found		
MountPoints2 (per user)	NTUSER\...\Explorer\MountPoints2	Found	12	
setupapi device install log	C:\Windows\INF\setupapi.dev*.log	Found	4	
Windows Search VolumeInfoCache	SOFTWARE\Microsoft\Windows Search\VolumeInfoCache	Found	7	
USB device history	Consolidated (registry + setupapi + event logs)	Found	2	
Cloud storage clients	OneDrive, Google Drive, Dropbox, Box, MEGA, iCloud, pCloud, Nextcloud	Found	9	
Outlook store	C:\Users\informant\AppData\Local\Microsoft\Outlook\iaman.informant@nist.gov.ost	Found	14	
Outlook PST/OST	*.pst / *.ost	Found	1	14 messages
Standalone e-mail files	*.eml / *.msg	Checked - nothing found		
Windows Search index	C:\ProgramData\Microsoft\Search\Data\Applications\Windows\Windows.edb	Found	464	
Sticky Notes	StickyNotes.snt / plum.sqlite	Found	1	
Documents (metadata + text)	user profiles + non-system volumes	Found	90	1,255 candidate files <= 50 MB
System information	SOFTWARE / SYSTEM hives	Found	19	
Local accounts	SAM\Domains\Account\Users	Found	6	
Network profiles	SOFTWARE\...\NetworkList\Profiles	Found	2	
Network interfaces	SYSTEM\...\Tcpip\Parameters\Interfaces	Found	4	
Installed programs	Uninstall keys (machine + user)	Found	66	
Remote access tool footprint	23 tools checked (files, services, installs, execution)	Checked - nothing found		

Artifact	Location checked	Result	Records	Notes
Remote access connection logs	AnyDesk / TeamViewer / ScreenConnect / RustDesk / Splashtop logs	Checked - nothing found		
Physical search	Vol1 (NTFS, 0.10 GB)	Checked - nothing found		full volume
File carving	Vol1 unallocated space	Found	1	3 signature candidates
Physical search	C: (NTFS, 21.37 GB)	Found	2764	full volume
File carving	C: unallocated space	Found	1	11,945 signature candidates
Physical search	Unpartitioned space (1.0 MB)	Checked - nothing found		
Physical search	Unpartitioned space (1.0 MB)	Checked - nothing found		
Document text search	full-text index of documents (incl. carved)	Checked - nothing found		
Memory file strings	C:\hiberfil.sys	Found	5	header zeroed (resumed / cleared hibernation file) - remnants searched
Memory file strings	C:\pagefile.sys	Found	16	
Target file search	all locations	Skipped		no target files / hashes supplied
Raw artifact collection	C:\Users\sujay\Desktop\Forensic Cases\CFReDS-DataLeakage\Collected\E01_Suspect workstation	Found	655	392 MB copied
CD/DVD burn staging area	Users*\AppData\Local\Microsoft\Windows\Burn\Burn* (MFT + \$UsnJrnl)	Found	62	8 IMAPI mastering session(s) in \$UsnJrnl
CD Burning registry	NTUSER\Software\Microsoft\Windows\CurrentVersion\Explorer\CD Burning	Found	7	
Run / RunOnce keys	HKLM + HKCU ... \CurrentVersion\Run*	Found	5	
Winlogon / IFEO / Applnit / LSA	SOFTWARE\...\Winlogon, IFEO, Applnit_DLLs, Control\Lsa	Checked - nothing found		
Services	SYSTEM\CurrentControlSet\Services	Found	1522	
Scheduled tasks	C:\Windows\System32\Tasks	Found	41	
Startup folders	...\Start Menu\Programs\Startup	Checked - nothing found		
WMI subscriptions	wbem\Repository (CIM: __FilterToConsumerBinding)	Checked - nothing found		1 default Windows subscription(s) not reported

9. Limitations and assumptions

- Suspect workstation (INFORMANT-PC): the image hash was not re-verified in this run (container integrity checks only).
- Registry key last-write times record the last change to a key, not necessarily the event of interest; MRU lists only time-stamp their most recent entry.
- NTFS last-access timestamps may be disabled or updated lazily by Windows; they are used as supporting indicators only.
- Browser history shows pages visited; it does not by itself prove that a file was uploaded. Upload conclusions rely on corroboration (file dialogs, memory artifacts, DLP alerts).
- Absence of an artifact does not prove that an action did not take place - artifacts can be overwritten, rotated or deliberately removed.

Appendix A - Exported and carved files

Evidence	Source	Saved as	Size	SHA-256
Suspect workstation	carved Vol1 @ 0x2106000	Vol1_000002106000.exe	89168	9645c44d999e3983f0f7466dc0bafa26551f0495694c2f5653873196a4ff81f1
Suspect workstation	carved C: @ 0x25e5000	C_0000025E5000.exe	9512	e138050a76a472f0119dda67f02b2a6704525e91e53a8a86dcddcc60e47731254

Appendix B - Parsed artifacts and raw collection for manual analysis

Every artifact parsed by the tool is available as CSV for independent review (open in Excel / Timeline Explorer), and the original artifact files were copied out of the image with their hashes. Folder paths are relative to the case folder

C:\Users\sujay\Desktop\Forensic Cases\CFReDS-DataLeakage.

Suspect workstation (INFORMANT-PC)

Parsed file	Rows	Size
Parsed\E01_Suspect workstation\Coverage.csv	109	9 KB
Parsed\E01_Suspect workstation\SuperTimeline.csv	11143	2,806 KB
Parsed\E01_Suspect workstation\Anti-Forensics\Timestamp Anomalies (timestamp indicators).csv	3000	1,230 KB
Parsed\E01_Suspect workstation\Browser Activity\Browser Extensions.csv	22	9 KB
Parsed\E01_Suspect workstation\Browser Activity\Cookie Hosts (sessions with sites).csv	83	27 KB
Parsed\E01_Suspect workstation\Browser Activity\Downloaded Files (Zone.Identifier).csv	18	2 KB
Parsed\E01_Suspect workstation\Browser Activity\Typed URLs (IE _ Edge legacy _ Explorer).csv	5	1 KB
Parsed\E01_Suspect workstation\Browser Activity\Web Downloads.csv	4	4 KB
Parsed\E01_Suspect workstation\Browser Activity\Web History.csv	402	245 KB
Parsed\E01_Suspect workstation\Browser Activity\Web Search Terms.csv	29	13 KB
Parsed\E01_Suspect workstation\Cloud & Sync\Cloud Storage Accounts.csv	1	1 KB
Parsed\E01_Suspect workstation\Cloud & Sync\Cloud Synced Items.csv	8	3 KB
Parsed\E01_Suspect workstation\Collection\Collected Raw Artifact Files.csv	655	331 KB
Parsed\E01_Suspect workstation\Documents\Document Metadata.csv	3	3 KB
Parsed\E01_Suspect workstation\Documents\Sticky Notes.csv	5	1 KB
Parsed\E01_Suspect workstation>Email\E-mail Messages.csv	25	7 KB
Parsed\E01_Suspect workstation\Event Logs\Event Log Files.csv	54	17 KB
Parsed\E01_Suspect workstation\Event Logs - all records\Application.csv	2371	316 KB
Parsed\E01_Suspect workstation\Event Logs - all records\Microsoft-Windows-Application-Experience%4Program-Inventory.csv	68	37 KB
Parsed\E01_Suspect workstation\Event Logs - all records\Microsoft-Windows-Application-Experience%4Program-Telemetry.csv	2	0 KB
Parsed\E01_Suspect workstation\Event Logs - all records\Microsoft-Windows-Bits-Client%4Operational.csv	178	73 KB
Parsed\E01_Suspect workstation\Event Logs - all records\Microsoft-Windows-BranchCacheSMB%4Operational.csv	27	15 KB
Parsed\E01_Suspect workstation\Event Logs - all records\Microsoft-Windows-Dhcp-Client%4Admin.csv	1	0 KB
Parsed\E01_Suspect workstation\Event Logs - all records\Microsoft-Windows-Dhcpv6-Client%4Admin.csv	4	1 KB
Parsed\E01_Suspect workstation\Event Logs - all records\Microsoft-Windows-Diagnosis-DPS%4Operational.csv	48	20 KB
Parsed\E01_Suspect workstation\Event Logs - all records\Microsoft-Windows-Diagnosis-Scheduled%4Operational.csv	6	1 KB
Parsed\E01_Suspect workstation\Event Logs - all records\Microsoft-Windows-Diagnosis-Scripted%4Admin.csv	1	0 KB
Parsed\E01_Suspect workstation\Event Logs - all records\Microsoft-Windows-Diagnosis-Scripted%4Operational.csv	4	1 KB
Parsed\E01_Suspect workstation\Event Logs - all records\Microsoft-Windows-Diagnostics-Performance%4Operational.csv	28	20 KB
Parsed\E01_Suspect workstation\Event Logs - all records\Microsoft-Windows-GroupPolicy%4Operational.csv	395	72 KB
Parsed\E01_Suspect workstation\Event Logs - all records\Microsoft-Windows-Kernel-WHEA%4Operational.csv	8	68 KB
Parsed\E01_Suspect workstation\Event Logs - all records\Microsoft-Windows-Known Folders API Service.csv	46	12 KB
Parsed\E01_Suspect workstation\Event Logs - all records\Microsoft-Windows-LanguagePackSetup%4Operational.csv	2	0 KB
Parsed\E01_Suspect workstation\Event Logs - all records\Microsoft-Windows-MUI%4Operational.csv	14	3 KB
Parsed\E01_Suspect workstation\Event Logs - all records\Microsoft-Windows-NetworkLocationWizard%4Operational.csv	1	0 KB
Parsed\E01_Suspect workstation\Event Logs - all records\Microsoft-Windows-NetworkProfile%4Operational.csv	74	18 KB
Parsed\E01_Suspect workstation\Event Logs - all records\Microsoft-Windows-OfflineFiles%4Operational.csv	28	6 KB
Parsed\E01_Suspect workstation\Event Logs - all records\Microsoft-Windows-PrintService%4Admin.csv	3	1 KB
Parsed\E01_Suspect workstation\Event Logs - all records\Microsoft-Windows-ReadyBoost%4Operational.csv	26	6 KB
Parsed\E01_Suspect workstation\Event Logs - all records\Microsoft-Windows-Resource-Exhaustion-Detector%4Operational.csv	15	3 KB
Parsed\E01_Suspect workstation\Event Logs - all records\Microsoft-Windows-TerminalServices-LocalSessionManager%4Operational.csv	35	11 KB
Parsed\E01_Suspect workstation\Event Logs - all records\Microsoft-Windows-User Profile Service%4Operational.csv	60	13 KB
Parsed\E01_Suspect workstation\Event Logs - all records\Microsoft-Windows-Windows Defender%4WHC.csv	24	4 KB

Parsed file	Rows	Size
Parsed\E01_Suspect workstation\Event Logs - all records\Microsoft-Windows-Windows Firewall With Advanced Security%4Firewall.csv	126	63 KB
Parsed\E01_Suspect workstation\Event Logs - all records\Microsoft-Windows-WindowsBackup%4ActionCenter.csv	4	1 KB
Parsed\E01_Suspect workstation\Event Logs - all records\Microsoft-Windows-WindowsSystemAssessmentTool%4Operational.csv	126	33 KB
Parsed\E01_Suspect workstation\Event Logs - all records\Microsoft-Windows-WindowsUpdateClient%4Operational.csv	67	10 KB
Parsed\E01_Suspect workstation\Event Logs - all records\OALerts.csv	50	1 KB
Parsed\E01_Suspect workstation\Event Logs - all records\Security.csv	2172	512 KB
Parsed\E01_Suspect workstation\Event Logs - all records\Setup.csv	39	11 KB
Parsed\E01_Suspect workstation\Event Logs - all records\System.csv	1642	437 KB
Parsed\E01_Suspect workstation\File & Folder Access\Explorer Search Terms (WordWheelQuery).csv	1	0 KB
Parsed\E01_Suspect workstation\File & Folder Access\Files Opened (IE _ WinInet file _ history).csv	15	7 KB
Parsed\E01_Suspect workstation\File & Folder Access\Jump Lists.csv	104	79 KB
Parsed\E01_Suspect workstation\File & Folder Access\Last Visited Folder MRU (by application).csv	3	1 KB
Parsed\E01_Suspect workstation\File & Folder Access\Microsoft Office File MRU.csv	10	3 KB
Parsed\E01_Suspect workstation\File & Folder Access\Open _ Save Dialog MRU.csv	10	4 KB
Parsed\E01_Suspect workstation\File & Folder Access\Recent Documents (RecentDocs).csv	34	12 KB
Parsed\E01_Suspect workstation\File & Folder Access\Recycle Bin.csv	14	8 KB
Parsed\E01_Suspect workstation\File & Folder Access\Shellbags (folders browsed).csv	113	37 KB
Parsed\E01_Suspect workstation\File & Folder Access\Shortcut (LNK) Files.csv	226	151 KB
Parsed\E01_Suspect workstation\File System\FileSystem_C_NTFS.csv	77822	33,284 KB
Parsed\E01_Suspect workstation\File System\FileSystem_Vol1_NTFS.csv	87	26 KB
Parsed\E01_Suspect workstation\File System\USN Journal.csv	1	0 KB
Parsed\E01_Suspect workstation\File System\UsnJrnl_J.csv	317137	62,018 KB
Parsed\E01_Suspect workstation\File System\Volumes.csv	2	0 KB
Parsed\E01_Suspect workstation\Malware\Scripts _ Containers in User-Writable Folders.csv	474	358 KB
Parsed\E01_Suspect workstation\Memory Artifacts\Memory File Strings (pagefile _ hiberfil).csv	21	5 KB
Parsed\E01_Suspect workstation\Network\Mapped Network Drives.csv	1	0 KB
Parsed\E01_Suspect workstation\Network\Network Events.csv	145	135 KB
Parsed\E01_Suspect workstation\Network\Network Interfaces.csv	4	1 KB
Parsed\E01_Suspect workstation\Network\Network Profiles (Wi-Fi _ LAN).csv	2	1 KB
Parsed\E01_Suspect workstation\Network\Network Settings (hosts _ proxy _ VPN).csv	1	0 KB
Parsed\E01_Suspect workstation\Persistence\Autostart Entries (Run keys _ Winlogon _ IFEO _ Startup).csv	5	2 KB
Parsed\E01_Suspect workstation\Persistence\Persistence _ Configuration Events.csv	56	96 KB
Parsed\E01_Suspect workstation\Persistence\Scheduled Tasks.csv	41	21 KB
Parsed\E01_Suspect workstation\Persistence\Service Events.csv	89	39 KB
Parsed\E01_Suspect workstation\Persistence\Services.csv	1522	519 KB
Parsed\E01_Suspect workstation\Persistence\WMI Event Subscriptions.csv	4	2 KB
Parsed\E01_Suspect workstation\Program Execution\Application Events.csv	184	105 KB
Parsed\E01_Suspect workstation\Program Execution\MUICache (executed program display names).csv	5	1 KB
Parsed\E01_Suspect workstation\Program Execution\Prefetch - files referenced.csv	2960	1,266 KB
Parsed\E01_Suspect workstation\Program Execution\Prefetch.csv	95	643 KB
Parsed\E01_Suspect workstation\Program Execution\Run Box History (RunMRU _ Win+R).csv	2	1 KB
Parsed\E01_Suspect workstation\Program Execution\Shimcache (AppCompatCache).csv	305	89 KB
Parsed\E01_Suspect workstation\Program Execution\UserAssist (GUI program execution).csv	105	29 KB
Parsed\E01_Suspect workstation\Remote Access\Remote Desktop _ Remote Session Events.csv	35	21 KB
Parsed\E01_Suspect workstation\Search\Carved Files (unallocated space).csv	2	1 KB
Parsed\E01_Suspect workstation\Search\Keyword Search Summary.csv	3	4 KB
Parsed\E01_Suspect workstation\Search\Keyword _ Hits.csv	1152	766 KB
Parsed\E01_Suspect workstation\System Information\Installed Programs.csv	66	30 KB
Parsed\E01_Suspect workstation\System Information\System Information.csv	16	2 KB
Parsed\E01_Suspect workstation\System Information\System Power _ Time Events.csv	79	46 KB
Parsed\E01_Suspect workstation\USB & Removable Media\CD Burning Registry.csv	12	3 KB
Parsed\E01_Suspect workstation\USB & Removable Media\Files Staged for CD _ DVD Burning.csv	31	10 KB
Parsed\E01_Suspect workstation\USB & Removable Media\MountedDevices.csv	10	3 KB
Parsed\E01_Suspect workstation\USB & Removable Media\Optical Disc Write Events.csv	4	2 KB
Parsed\E01_Suspect workstation\USB & Removable Media\USB Connection Sessions.csv	2	1 KB
Parsed\E01_Suspect workstation\USB & Removable Media\USB _ Device Events.csv	33	27 KB
Parsed\E01_Suspect workstation\USB & Removable Media\USB _ Removable Devices.csv	2	4 KB
Parsed\E01_Suspect workstation\USB & Removable Media\User Mounted Volumes (MountPoints2).csv	12	3 KB
Parsed\E01_Suspect workstation\USB & Removable Media\Windows Search VolumeInfoCache.csv	7	2 KB
Parsed\E01_Suspect workstation\Users & Accounts\Account Management Events.csv	62	54 KB
Parsed\E01_Suspect workstation\Users & Accounts\Local User Accounts.csv	6	2 KB
Parsed\E01_Suspect workstation\Users & Accounts\Logon _ Logoff Events.csv	309	253 KB
Parsed\E01_Suspect workstation\Windows Search\Windows Search Index Items.csv	464	256 KB

Raw collection: 655 files, 578.5 MB in Collected\E01_Suspect workstation (hash manifest: _manifest.csv).

Group	Files	Size
Browsers	172	107.6 MB
Cloud	8	0.4 MB

Group	Files	Size
Devices	3	4.1 MB
E-mail	19	43.4 MB
Event logs	54	10.9 MB
Execution	185	7.8 MB
File access	74	14.9 MB
NTFS metadata	7	186.2 MB
Registry	120	141.6 MB
Search index	1	42.0 MB
Startup	4	0.0 MB
Sticky Notes	1	0.0 MB
System	7	19.5 MB

Appendix C - Software

Windows Forensic Automation	1.2.0
Python	3.11.9
dissect.target	
python-docx	1.2.0
Qt (PySide6)	6.11.2
Host	Windows 10
libewf	20260924

Appendix D - Glossary of artifacts

- **MFT (\$MFT)**: NTFS Master File Table; one record per file, including records of deleted files until they are reused.
- **\$STANDARD_INFORMATION / \$FILE_NAME**: Two sets of NTFS timestamps; discrepancies between them can indicate timestamp manipulation.
- **USN journal (\$UsnJrnl:\$J)**: NTFS change journal recording file creation, modification, renaming and deletion.
- **File slack**: Space between the end of a file and the end of its last cluster; may hold remnants of earlier data.
- **Unallocated space**: Clusters not assigned to any file; deleted file content remains here until overwritten.
- **pagefile.sys / hiberfil.sys / swapfile.sys**: Memory written to disk by Windows; retains fragments of documents, URLs and commands.
- **Volume Shadow Copy**: Point-in-time snapshot of a volume kept by Windows; can contain earlier versions of deleted files.
- **LNK (shortcut) file**: Created when a user opens a file; records the target path, its timestamps, size, drive type, volume serial number and volume label.
- **Jump list**: Per-application list of recently / frequently opened files with access times and counts.
- **Shellbags**: Registry records of folders viewed in Explorer, including folders on removable and network drives.
- **RecentDocs / OpenSavePidlMRU / LastVisitedPidlMRU**: Registry lists of recently opened documents, files chosen in Open/Save dialogs and the folder each application last used in such a dialog.
- **RunMRU**: Commands typed into the Windows Run (Win+R) box - the primary artifact for ClickFix / paste-and-run attacks.
- **UserAssist**: Registry record of programs started through the Explorer GUI, with run counts and last run time.
- **Prefetch**: Files created when a program runs; record up to 8 run times and the files the program loaded.
- **Amcache / Shimcache / BAM**: Program inventory and compatibility caches that record executables present on or run by the system.
- **USBSTOR / MountedDevices / MountPoints2**: Registry records of USB storage devices, the drive letters / volumes they were mounted as, and the user profiles that mounted them.
- **Partition/Diagnostic event 1006**: Windows 10+ event logged at every disk arrival and removal, including the device serial and the first sector of each volume (volume serial number and label).
- **setupapi.dev.log**: Driver installation log recording the first time each device was installed.

- **Zone.Identifier:** Mark-of-the-Web alternate data stream recording the URL a file was downloaded from.
- **Windows Timeline (ActivitiesCache.db):** Database of applications and documents used, with focus start and end times.
- **SRUM:** System Resource Usage Monitor; per-application network bytes sent / received in hourly intervals.
- **Carving:** Recovery of files from unallocated space by recognizing file signatures and structures.

Examiner declaration

I confirm that the examination described in this report was carried out by me or under my supervision, that the evidence was handled in a manner that preserved its integrity, and that the findings are a true reflection of the artifacts examined.

Examiner	Examiner
Signature	
Date	